

Foreword

Dear valued user,

Thank you for your trust and support in our factory's products! The HWL series mini excavator, bearing the sykend brand, is a type of excavator that focuses on excavation operations with auxiliary lifting and loading capabilities. It features a compact design, appropriate power matching, good product stability, and high cost-effectiveness. It can meet the excavation and loading requirements in different working conditions such as plains, hills, and forest areas. Additionally, it is suitable for various applications, including brick factories, kiln plants, waterways, construction, dredging, and road construction, making it the ideal construction machinery for agricultural operations and small-scale projects.

To ensure users have a correct understanding of the machine's usage, adjustment, maintenance, and upkeep, and to fully utilize the excavator's capabilities, please carefully read this Operation and Maintenance Manual and strictly follow the instructions provided. For information about the usage and maintenance of the matched engine, please refer to the "Engine User Manual" prepared by the engine manufacturer.

The operation section serves as a technical reference for operators, providing graphical instructions to guide them in correctly inspecting, starting, operating, and stopping the machine. The operational techniques outlined in the manual serve as a foundation for operators to improve their skills and expertise by gaining knowledge about the machine and its functions.

The maintenance section serves as a guidebook for users to perform whole-machine maintenance. The demonstration images are arranged based on the maintenance cycle, from short to long intervals. Specific maintenance measures for the machine are detailed in the "Lubrication and Maintenance Schedule." Users should perform maintenance according to the requirements specified in the "Lubrication and Maintenance Schedule," adjusting the maintenance items based on the machine's operating hours. In extremely harsh, dusty, or humid working conditions, lubrication frequency may need to be increased based on machine operation.

To better illustrate certain structural features of the machine, some demonstration images in this manual are shown as perspective views, and thus may differ from the actual product's

appearance. As technical improvements may lead to changes in the actual vehicle's mechanical structure, technical parameters, etc., which are not shown in this manual, please consult our company for the latest product information.

Before using or conducting maintenance on the machine, relevant documentation should be reviewed, and if necessary, contact our technical service station. When purchasing parts, please provide the excavator's production date and serial number.

To facilitate your usage, we have established a comprehensive "Three Guarantees Technical Service Network." If, during the warranty period, any quality issues arise with the purchased HWL series mini excavator under proper use, please contact the local HWL excavator warranty maintenance service station promptly.

As our company's product technology undergoes continuous innovation, we reserve the right to interpret and modify this manual. If there are discrepancies between the actual product and the images in this manual, the actual product shall prevail.

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Safety

General Precautions

You have a responsibility to comply with safety regulations and laws from relevant authorities and operate, inspect, and maintain the machine according to the manufacturer's requirements. In fact, all accidents occur due to failure to adhere to basic safety regulations and precautions. Most accidents can be avoided by identifying potential hazards in advance. Please read and understand all safety information on how to prevent accidents, and do not operate the machine until you are confident in how to operate, inspect, and maintain it correctly.

Compliance with all safety regulations:

The machine must be operated, inspected, and maintained by trained and qualified personnel. During machine operation, inspection, and maintenance, it is essential to understand and comply with all rules, regulations, precautions, and safety measures. Do not operate, inspect, or maintain the machine under the influence of alcohol, drugs, medication, fatigue, or lack of sleep.

When detecting abnormal conditions in the machine:

During machine operation, inspection, or maintenance, if any abnormalities (noise, vibration, unusual odor, oil leakage, false alarms, etc.) are detected, inform the sales or service agent immediately and take appropriate actions. Do not operate the machine until the abnormalities are resolved.

Operating temperature range:

To maintain machine performance and avoid premature wear, please follow the operating conditions below:

Do not operate the machine if the outdoor temperature exceeds +45°C or falls below -15°C.

Operating the machine at temperatures above +45°C may cause engine overheating, leading to a decrease in engine oil performance. Additionally, hydraulic oil may become very hot, causing damage to hydraulic components.

Operating the machine at temperatures below -15°C may cause rubber components like gaskets to become hard, resulting in premature wear or damage to the machine.

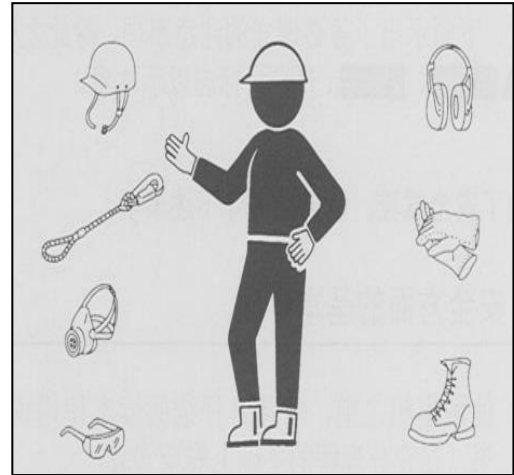
If you need to operate the machine outside the above-mentioned temperature range, consult your sales or service agent.

Wear appropriate clothing and protective equipment:

Do not wear loose clothing or adornments that could get caught on control levers or moving parts. Avoid wearing clothing with oil or fuel stains that could easily catch fire.

Wear safety shoes, helmets, safety goggles, respirators, thick gloves, ear protection, and other protective equipment based on the work environment requirements. When using a breaker or compressed air, wear suitable protective equipment such as safety goggles and respirators, as metal chips or other objects may cause serious injuries.

Use hearing protection devices when operating the machine. Prolonged exposure to high noise levels can cause hearing impairment or even complete loss of hearing.



Installing Fire Extinguishers and First Aid Kits

Be prepared for fires and accidents.

Install fire extinguishers and first aid kits and learn how to use them.

Learn how to extinguish fires and handle accidents.

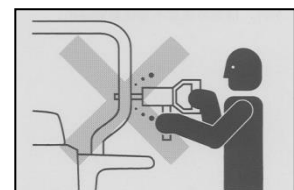
Know how to contact emergency assistance and create an emergency contact list.



Don't remove safety devices

Ensure that all protective guards and barriers are properly installed and secure. Before operating the machine, repair or replace any damaged parts.

Do not remove any safety devices except during maintenance. Keep all safety devices in good working condition at all times.



Equip signalmen and flaggers

Learn how to use specific hand signals required for the job and identify the person responsible for giving the signals.

All personnel must fully understand all hand signals.

Operators must respond only to the signals of designated personnel but



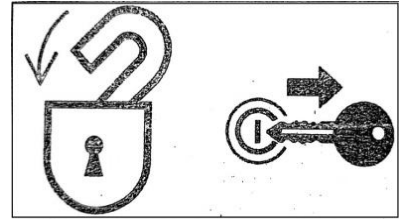
must obey any stop signals from anyone at any time.

Signalmen must stand in a clear and visible location when giving signals.

Precautions when standing up or leaving the operator's seat

When standing up from the operator's seat, lower the working equipment to the ground and stop the engine. Accidentally touching any control lever can cause the machine to move and result in serious injury or death.

Please note that you should not touch the blade, boom swing, and auxiliary hydraulic controls. Before leaving the operator's seat, lower the working equipment to the ground and turn off the engine. Also, make sure to remove the key and carry it with you, then store it in the designated place.



Avoiding Fire and Explosion Hazards

Keep fuel, lubricating oil, grease, and antifreeze away from ignition sources. Fuel is highly flammable and extremely dangerous.

When handling these flammable substances, stay away from lit cigarettes, matches, lighters, and other flames or ignition sources.



Do not smoke or have an open flame while handling fuel or performing operations related to the fuel system.

Do not leave the work area while refueling or adding lubricating oil.

Do not remove the fuel tank cap or refuel while the engine is running or has not cooled down. Also, do not splash fuel onto the machine's hot surfaces or electronic system components.

Immediately clean up any spilled fuel or lubricating oil.

Check for fuel and lubricating oil leaks, and eliminate any leaks and clean the machine before operating it.

When polishing or welding, move flammable items to a safe location.

Do not cut or weld on pipes or tubes containing flammable liquids. Before cutting or welding, thoroughly clean with non-flammable solvents.

Clear all debris or debris from the machine, ensuring that no oily rags or other flammable materials are left on the machine.

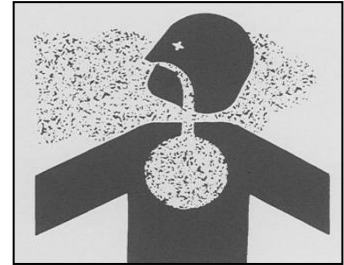
Handle various solvents or dry chemicals (foam fire extinguishers) according to the procedures on the manufacturer's containers, in well-ventilated areas.

Do not use fuel for cleaning purposes; always use non-flammable solvents.

Store all flammable liquids and materials in a safe and well-ventilated place.

Short circuits in the electronic system can cause fires. Check daily for loose or damaged wire connections, tighten loose connectors and cable clips, and repair or replace damaged wires.

Fires caused by pipes: Ensure that hose and pipe clamps, protective devices, and cushions are securely fastened. Loose hoses or pipes may be damaged during operation due to vibration or contact with other components, which can result in high-pressure oil spraying, causing fire or injury.



Engine exhaust gases are toxic

Do not operate the engine in poorly ventilated enclosed areas. If natural ventilation is not possible, install exhaust fans, fans, extended exhaust pipes, or other ventilation devices.

Handling Asbestos Dust

Inhaling asbestos dust can cause lung cancer. When handling materials that may contain asbestos, take the following safety measures:

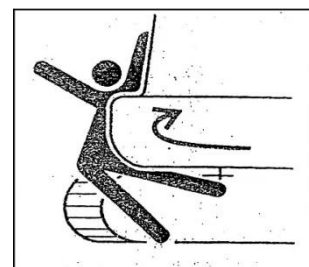
1. Do not use compressed air for cleaning.
2. Avoid polishing or grinding parts containing asbestos.
3. When cleaning, use vacuum equipment with high-efficiency particulate air (HEPA) filters.
4. If other methods of dust control are not available, wear the specified respirator. When working indoors, install a ventilation system with high-efficiency particulate filters.

During operations, do not allow unauthorized personnel to enter the work area.

Please strictly comply with the rules and environmental standards applicable to the work area.

Be cautious to avoid crushing injuries

Do not place hands, feet, or other body parts between the steps and tracks, between the body and working equipment, or between the cylinders and moving parts. The size of these gaps may change when the machine is in motion, potentially causing serious injuries or death.



Use optional products:

Before installing optional products, consult with sykend. Depending on the type or combination of attachments, they may come into contact with the operator's cab or other parts of the machine. Before use, ensure that the installed options do not come into contact with other parts.

Do not use attachments that are not approved by sykend, as doing so may

jeopardize safety or adversely affect the machine's operation or service life. sykend takes no responsibility for any injuries, accidents, or product damages caused by the use of unauthorized attachments.

Do not modify the machine:

Unauthorized modifications to the machine can cause injuries or death. Do not make any unauthorized modifications to any part of the machine.

Safety signs (labels):

To ensure the safety of operators and personnel around the work area, the following safety signs (labels) should be placed in certain parts of the machine. Walk around the machine with this manual and review the content and placement of these safety signs. Review these signs and operating instructions in this manual with your machine operator.

Safety signs should be kept clean, clear, and easily recognizable. If any safety labels become detached or damaged and are no longer legible, replace them with new labels. When ordering new labels from the sykend service agent, provide your product serial number.

When the parts/body where safety signs are posted have been replaced, apply new labels to the new parts / body.

Precautions during Preparation

Understand the work area

Before starting the operation, it is essential to understand the work area to ensure safety.

Check the terrain and ground conditions of the work area, or in the case of indoor operations, inspect the building structures and take necessary safety measures if required.

Avoid any hazards and obstacles, such as ditches, underground pipelines, trees, cliffs, overhead power lines, falling rocks, or areas prone to landslides.

Coordinate with the site manager to identify the locations of buried gas pipes, water pipes, and power cables. If necessary, consult with the manager and establish specific safety measures to ensure safety.

Ensure the safety of pedestrians and vehicles when working on roads.

Utilize signal persons and/or signals. Isolate the work area to prevent unauthorized personnel from entering.

When working in water or crossing shallow streams, check water depth, ground firmness, and water flow speed in advance.

Check the strength of the bridge.

When walking on a bridge or structure, check the permissible load. If the strength is insufficient, reinforce the bridge or structure.

Always keep the machine clean.

Wipe away lubricants, grease, mud, snow, or ice to prevent accidents due to slipping.

Remove all loose objects and unnecessary devices from the machine.

Clean dust, lubricants, or grease from the engine area to prevent fires.

Clean around the operator's seat and remove any unnecessary objects from the machine.

Perform daily checks and maintenance.

Failure to identify or repair abnormal conditions or damage to the machine can lead to accidents.

Before operation, perform the specified checks and make necessary repairs immediately.

In case of an accident or engine failure where the machine cannot be operated, follow the shutdown procedure immediately and securely park the machine until the issue is fixed.



Precautions inside the operator's cab.

Before entering the operator's cab, clean the soles of your shoes from mud and grease. Operating the machine pedals with muddy or greasy shoes can cause accidents due to slipping.

Do not place parts or tools around the driver's seat.

Do not place any plastic bottles or install any suction cups inside the cab. Plastic bottles or suction cups can act as lenses and may cause fires.

Precautions during startup

When getting on/off the machine, support your weight in a three-point contact stance. Do not jump on or off the machine, and do not attempt to board or dismount a moving machine.

When entering/exiting the operator's cab, first fully open the door and check that it is locked in place (for machines with cabs).

Face the machine, use the handrails, and step onto/off the footboard in a three-point

contact stance (hands and feet) to support your weight.

Do not use the control levers as handrails.

Before starting the machine, ensure that any unauthorized persons have left the area.

Only start the machine when you have checked the following items and are certain it is safe to do so.

Walk around the machine, warning maintenance personnel and pedestrians near the machine, and ensure no one is around before starting the machine.

Check for any "Do not operate" or similar warning signs on the driver's cab, controls, or starting switches. If present, do not start the engine or touch any controls.

Sound the horn to alert people around the machine.

Sit on the operator's seat and start the engine.

Adjust the seat to a secure position and lock it in place.

Fasten your seat belt.

Check if the parking brake is engaged, and all control levers and pedals are in the neutral position.

Ensure that no one is near the machine.

Start and operate the machine only from the operator's seat.

Do not attempt to start the engine by short-circuiting the starting terminal.

When using jump-start cables, follow the recommended procedure strictly. Improper use of jump-start cables can cause battery explosion or unexpected machine movements.



After starting the engine

After starting the engine, perform the following operations and checks in an area free of personnel or obstructions. If any faults are detected, shut down the machine following the procedure and report the issue.

Preheat the engine and hydraulic oil.

Check if all gauges and warning devices are working properly.

Inspect for any unusual noises.

Test the engine speed control.

Operate all controls to ensure their proper functioning.



For cold climates:

Be cautious as the ground, pedals, and handrails may be slippery due to freezing.

In extremely cold weather, do not touch any metal parts of the machine with bare hands, as the skin can freeze to the metal, causing severe injuries.

Do not use ether or starting fluid on the engine, as it can cause explosions and result in severe injuries or death.

Preheat the engine and hydraulic oil. Failure to preheat before operating the control levers may cause the machine to respond or move improperly, leading to accidents.



Safety Signs and Symbols

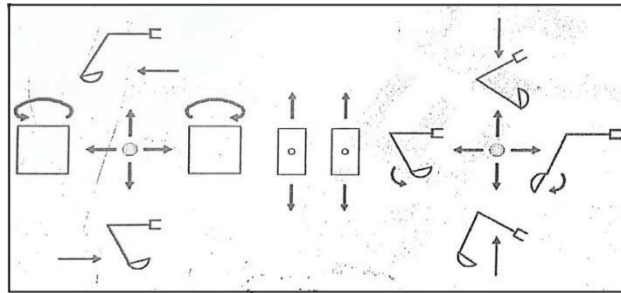
1. Work Equipment Operation

Instructions:

This sign is located on the inside front of the cabin.

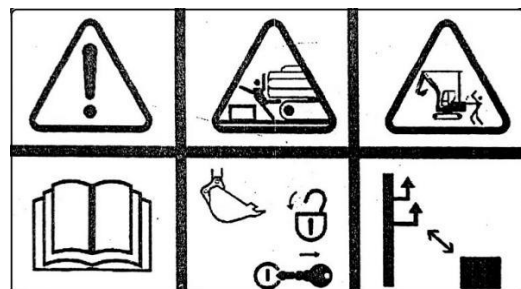
Check if the machine's control mode matches the label. If different, replace the label to match the machine's control mode before operating.

Failure to follow these instructions may result in serious injury or death.



2. Attention to Adjusting Track Tension:

3. Operating Warning, Moving and Working Warning, High Voltage Warning, Read Instruction Manual Warning, Hydraulic Lock Warning, Height Warning Signs.

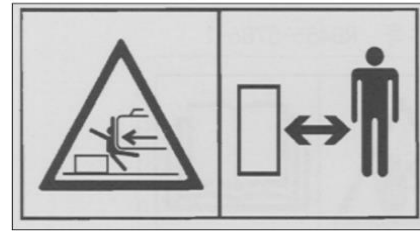


4. Keep Away from Swing Area Sign:

This sign is located on the rear counterweight.

Standing in the machine's swing area is prohibited.

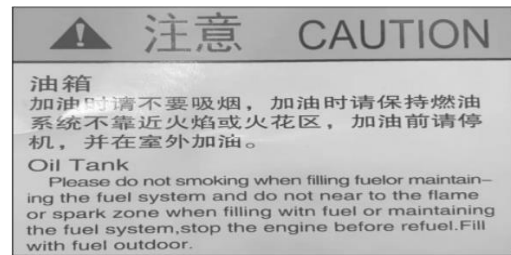
Do not damage or remove this sign from the machine.



5. Fuel logo:



6. Precautions for fuel:



7. Precautions for battery use



8. Keep out of the work area



9. Stay out of digging range



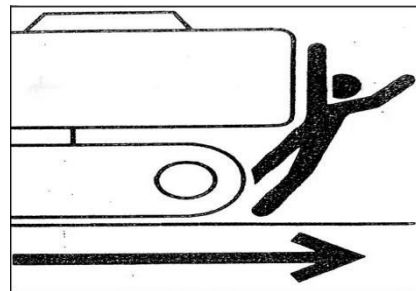
10. Hydraulic oil identification



Precautions during operation

Ensure good visibility:

1. When operating in dark areas, turn on the machine's work lights and headlights, and install additional lighting equipment if necessary.
2. In case of poor visibility due to adverse weather conditions (fog, snow, rain, or haze), stop operating the machine until visibility improves.
3. Never allow anyone to ride on any part of the machine while it is in motion or being operated.
4. Before operation, inspect the work area for safety and reliability and confirm the machine's performance limits.
5. Use a signaler in areas such as road shoulders, narrow spaces, or obstructed views. Do not allow anyone to enter the machine's swing radius and path.
6. Signal your intention to move by sounding the horn.
7. Be aware of blind spots behind the machine. Before walking backward, check the rear area for safety and ensure there are no people behind.



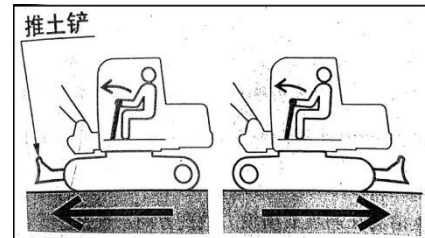
Ensure good visibility:

1. When operating in dark areas, turn on the machine's work lights and headlights, and install additional lighting equipment if necessary.
2. In adverse weather conditions (fog, snow, rain, or haze) causing poor visibility, stop operating the machine until visibility improves.
3. Never allow anyone to ride on or operate the machine while it is in motion. No one is permitted to be on any part of the machine during operation.
4. Before operation, inspect the work area for safety and reliability, and confirm the machine's performance limits.
5. Use a signaler in areas such as road shoulders, narrow spaces, or obstructed views. Do not allow anyone to enter the machine's swing radius and path.

6. Use the horn to signal your intention to move.
7. Be aware of blind spots behind the machine. Before walking backward, check the rear area for safety and ensure there are no people behind.

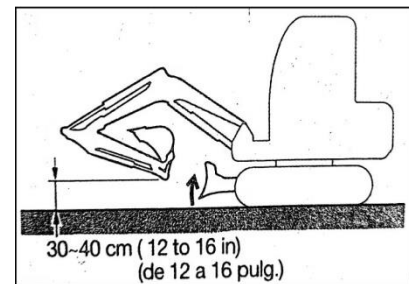
Before moving, check the position of the undercarriage (tracks).

Before operating the travel control lever/pedals, ensure that the bulldozer blade is located in front of the driver's seat. Please remember that when the bulldozer blade is located behind the driver's seat, the direction of operation of the travel control lever/pedals must be opposite to when the bulldozer blade is located in front of the driver's seat.



Safe Walking

When walking, ensure that the bulldozer blade is in the raised position. Fold up the blade attachment as shown in the right figure and raise the blade to a height of 30 to 40 centimeters above the ground. During walking, avoid making any rotations. If it is necessary to operate the blade attachment while walking, maintain a slow speed to ensure complete control at all times.



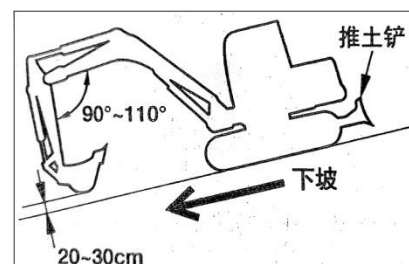
Try to avoid crossing obstacles. If it is unavoidable, keep the blade attachment close to the ground and walk slowly, avoiding obstacles that may cause the machine to tilt at an angle of 10° or greater.

On uneven terrain, maintain a low walking speed and avoid sudden acceleration, stops, or changes in direction. Otherwise, the work attachment may come into contact with the ground, causing the machine to lose balance and damage the surrounding structures.

Precautions for Walking on Slopes

When walking on slopes or inclines, pay attention to prevent the machine from overturning or sliding downhill.

Do not walk on slopes that are too steep for the machine to maintain its stability (maximum slope: 30°, lateral inclination: 10°). Please note that in

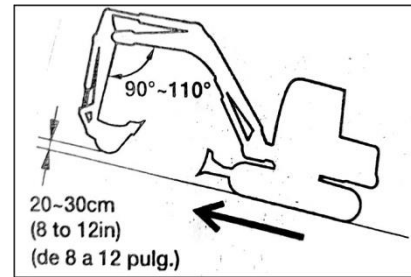


actual working conditions, the stability of the machine may be lower than the above values.

When climbing, keep the driver's seat facing the uphill; when descending, keep the driver's seat facing downhill. In both cases, pay attention to the ground in front of the machine while walking.

When walking on a slope, lower the blade to a height of 20 to 30 centimeters above the ground.

When climbing steep slopes, extend the blade attachment to the front. In emergencies, lower the blade to the ground and turn off the engine.



When walking on slopes or inclines, drive at a slow speed. Reduce the engine speed when going downhill.

Do not reverse downhill.

Do not change direction on slopes or side slopes. First, return to a flat surface and then choose another path.

On gentle slopes covered with grass or dry leaves, or on wet metal plates or frozen ground, the machine may slide sideways. Make sure the machine does not stop sideways on the slope.

If the machine stalls on a slope, return all controls to the neutral position, and then restart the engine.

Be extra cautious when operating the machine on snow or ice.

When walking on snowy or icy ground, drive at a low speed and avoid sudden acceleration, stops, or changes in direction.

In snowy areas, objects placed on the roadside are buried in snow, and the danger of the machine overturning or colliding with covered objects cannot be seen, so always operate with caution.

If the machine enters thick snow, there is a risk of overturning or getting stuck in the snow. Drive carefully and do not exceed the roadside or get trapped in the snow.

For frozen ground surfaces, when the temperature rises, the ground becomes soft, which may cause the machine to overturn and trap the operator inside.

When parking the machine on unstable ground, lower the dozer blade.

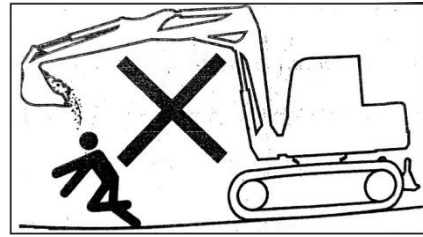
Do not move the bucket above personnel's head level.

Moving the bucket above personnel's head level may result in the scattering of the loaded material or the sudden dropping of the bucket, posing a danger.

Ensure driver's safety during loading.

Do not load until the driver reaches a safe location.

Do not swing or park the bucket above personnel or the cab. Load from behind the truck.

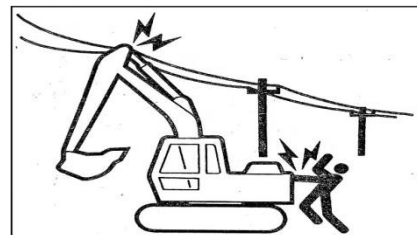


Maintain a safe distance from overhead high-voltage cables.

Do not bring any part of the machine or the loaded material close to high-voltage cables unless all safety measures required by local and national authorities have been taken. There is a risk of electric shock and death if personnel come close to a machine that is discharging sparks or is adjacent to or in contact with the power source. Always maintain a safe distance between the machine and the high-voltage cables.

Before starting operations, verify the procedures for safe operations with the local power company.

Treat all cables as high-voltage cables, even if it is known or believed that the power has been disconnected and the cables are visibly grounded. If the machine is too close to high-voltage lines, use signal persons to issue warnings.



Be aware that all personnel in the work area must not approach the machine or the loaded material.

Pay special attention to high-voltage cables buried underground.

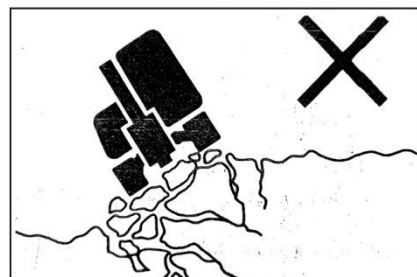
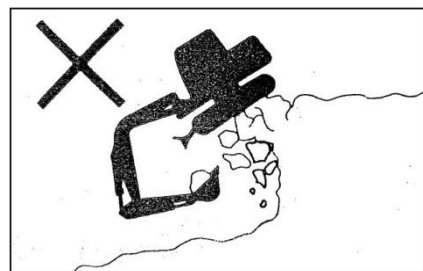
Pay close attention to hazardous operating conditions.

Do not excavate at the bottom of high embankments, as it may cause ground collapse and is therefore very dangerous.

Do not operate in areas with a risk of rockfall.

Maintain a safe distance between the machine and the edge of the excavation site; do not excavate the ground directly in front of the machine.

When working near cliffs or road shoulders, position the tracks perpendicular to the cliff or road shoulder and place the bucket in front for easy escape in case of any issues.



Avoid entering soft ground areas, as this may cause the machine to tilt or tip due to its weight, leading to overturning or sinking.

Do not approach unstable ground (cliffs, road shoulders, deep ditches). There is a risk of the ground collapsing due to the weight or vibration of the machine, which may result in the machine falling or tipping over.

Keep in mind that the soil may become unstable after heavy rain or blasting.

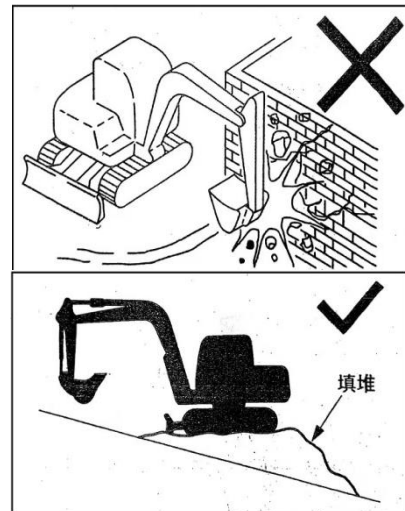
The ground on top of embankments and around excavated ditches is also unstable.

Do not conduct dismantling work under the machine, as the ground may become unstable, posing a risk of the machine falling.

Before working on top of buildings or other structures, check the strength and structure. The collapse of buildings or structures can cause serious injuries or damage.

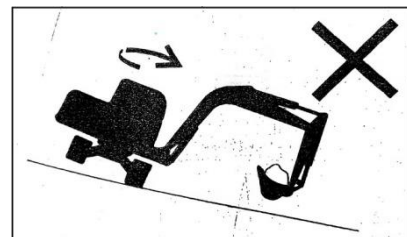
During dismantling work, do not work directly above; there is a risk of falling debris or building collapse, resulting in serious injuries or damage.

Do not use the impact force of the bucket for demolition work; there is a risk of serious injuries due to flying material fragments and damaged bucket operation equipment.



Operating on a slope is very dangerous.

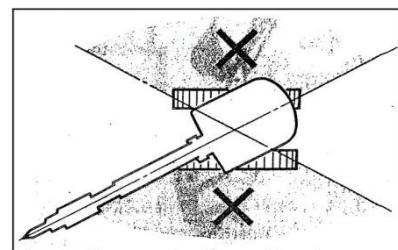
When operating on a slope or ramp, the rotation or operation of the working equipment may cause the machine to become unstable and overturn. It is recommended to avoid operating on slopes as much as possible.



Level the working area.

When the bucket is loaded with material, avoid rotating downhill as it can reduce the machine's stability and may cause it to overturn.

Do not swing the bucket when it has a heavy load, especially when operating on a slope. The machine is more prone to overturning in the lateral direction than in the longitudinal direction. When the bucket has a heavy load at the end of the working equipment, do not swing

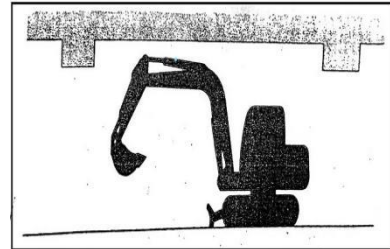


it laterally, especially on a slope.

For machines equipped with a breaker, crusher, or long arm, the attachment at the end is heavier compared to a standard bucket. For such machines, avoid digging or operating the boom (arm) downhill or towards the side.

Watch out for overhead objects.

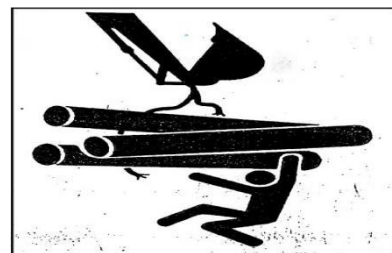
When operating under bridges, in tunnels, near cables, or indoors, be careful not to hit overhead objects with the boom or arm of the machine.



The excavator is not designed for lifting operations.

This machine is specifically designed for excavation work and does not have safety devices for lifting operations. If the excavator is used for lifting, extreme caution should be exercised.

Do not overload during lifting operations, as this can cause the machine to tip over, resulting in serious injury or death.



All rated lifting capacities are based on using the machine on stable and level ground. For safe lifting operations, users should allow for appropriate margins considering specific working conditions, including soft or uneven ground, non-horizontal positioning, lateral loads, dynamic or sudden loads, hazardous environments, and operator experience. Prior to operating the machine, the operator and other personnel should be thoroughly familiar with the operator's manual and strictly adhere to the equipment's safety procedures.

If chains or lifting devices are not correctly connected, the link or lifting device may fail, leading to serious injury or death.

When using this machine as a lifting crane, do not attempt to pull out tree stumps from the ground, as the safety of the load acting on the machine is unknown in this application.

Do not allow anyone to stand on or underneath the lifted object or approach the working area.

Beware of flying debris.

This machine is not equipped with protective equipment to shield the operator from flying debris. Do not use this machine in hazardous areas where the operator may be struck by flying debris.

Precautions for Towing:

During towing, improper operation, the use of incorrect cables, or inadequate inspection can lead to serious injury or death.

Use steel cables that match the towing force to prevent dangers in case of cable breakage or detachment.

Do not use twisted, entangled, or damaged cables.

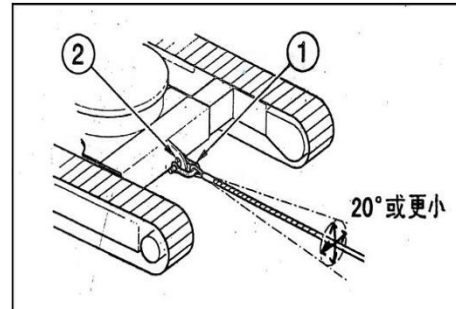
Avoid suddenly applying heavy loads on the cable.

When handling the cable, wear safety gloves.

Ensure that each machine involved in towing, whether the towing or towed machine, has its own operator.

Do not tow on slopes.

During towing, keep any personnel away from the cable.



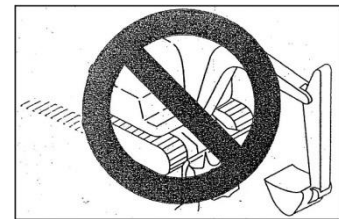
Prohibited operation

Do not operate on bedrock (hard or soft).

While walking, avoid rotating. If you must operate the bucket while walking, do it slowly to maintain full control at all times.

Do not use rotation force for dismantling or leveling operations.

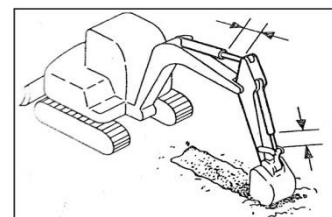
Do not use rotation force to dismantle walls or level the ground. Also, do not dig the bucket teeth into the ground during rotation, as it may damage the bucket mechanism.



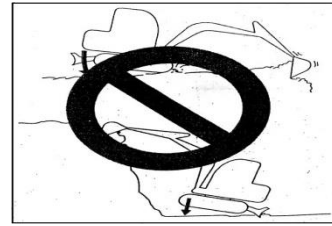
Do not dig while walking.

Avoid digging the bucket into the ground and using walking force for excavation.

Use hydraulic cylinders smoothly and do not fully extend them. Leave some room during operation.



When the machine body is lowered and the arm cylinder is fully extended, do not support the machine body using the bucket mechanism. This may overload the arm cylinder and cause damage.

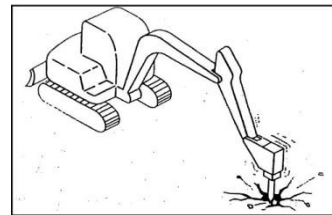


Do not use the sinking force of the machine during operations.

Applying additional tension on the machine will shorten its lifespan. During excavation, use the hydraulic pressure of the cylinders, keeping the depth shallow and the travel long.

Excavating Rock:

For hard rock excavation, it is advisable to use a hydraulic hammer or similar tool to break the rocks into smaller pieces before digging. This prevents damage to the machine and is more economical.



Pay attention to the impact on the bulldozer blade.

Striking rocks or other hard objects with the bulldozer blade can damage the blade and its hydraulic cylinders.

Be careful when retracting the bucket attachment.

When retracting the bucket, be mindful not to let it strike the bulldozer blade.

Do not use the bulldozer blade as a support.



Pay attention to the bulldozer blade during excavation.

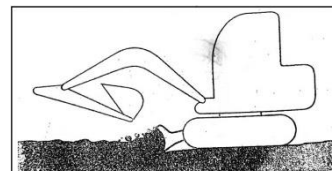
During deep excavation, if the bulldozer blade is in front, be cautious to avoid collisions between the arm, bucket, and the bulldozer blade.

Try to position the bulldozer blade at the rear during operations.



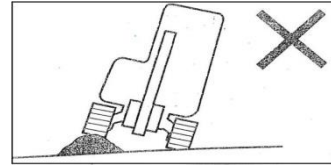
Be cautious when using the bulldozer blade for downward digging.

The bulldozer blade is designed for simple dozing operations, do not use it for deep digging, as this may damage the blade and the undercarriage.



Walking Precautions:

When walking over obstacles such as rocks or tree stumps, there may be significant loads applied to the machine, potentially causing damage. Avoid crossing obstacles if possible. If necessary, keep the bucket attachment close to the ground and proceed at a low speed, allowing the tracks to pass over the obstacle at the center.



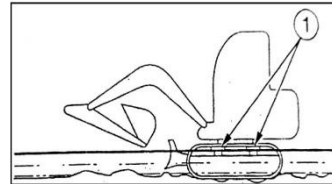
High-Speed Walking Precautions:

On uneven terrain, maintain low speed while walking to avoid sudden starts, stops, or changes in direction. Ensure the bulldozer blade is in front when walking at high speeds.



Using the Machine in Water:

Avoid immersing the rear part of the machine in water, as it may damage the radiator fan. The rear of the machine should not be submerged in water.



The water depth should not exceed three-quarters of the track height if using the machine in water.

For parts used in water for an extended period, sufficient grease should be applied until the old grease is pushed out.

Do not immerse the swing bearing or the main body in water or sand. If they become immersed, contact the sykend service agent for inspection.

To get the machine out of the mud, follow these steps:

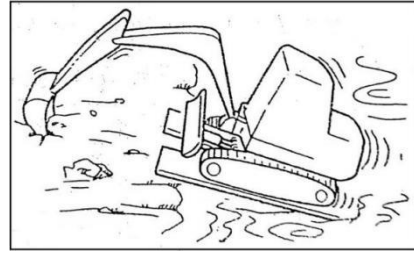
If one track is stuck in the mud:

1. Position the bucket towards the side of the stuck track.
2. Set the angle of the boom and arm to 90° to 110°.
3. Press the bottom of the bucket (not the bucket teeth) onto the ground.
4. Place a piece of wood or similar object under the lifted track.
5. Raise the bucket slowly to pull the machine out of the mud.



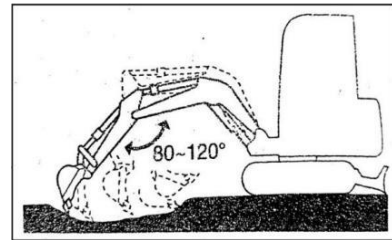
If both tracks are stuck in the mud:

1. Follow steps 1 to 4 for both tracks.
2. In front of the machine, dig the bucket into the ground.
3. Slowly walk forward while pulling the bucket lever to move the machine out of the mud.



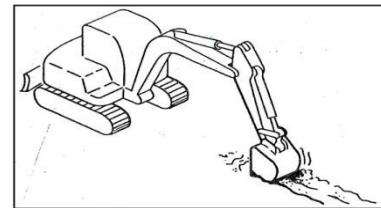
Excavation:

1. Place the bulldozer on the opposite side of the excavation area.
2. Use the boom and bucket to excavate. When excavating, keep the depth shallow and the stroke long. The maximum digging force is achieved when the angle between the boom and the arm is between 80° and 120° , so it is recommended to use this angle for effective excavation.



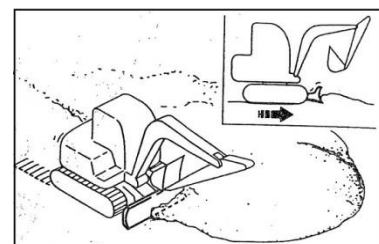
Excavating Ditches:

1. Install a bucket suitable for excavating ditches.
2. Adjust the tracks to be parallel to the ditch to be excavated for improved efficiency.
3. When excavating a wide ditch, start by digging the side, and then proceed to dig the middle. This sequence can enhance the effectiveness of the excavation process.



Leveling:

1. Position the bucket close to the machine body.
2. Slowly push the soil away from the side of the pile.
3. When the pile of soil is lower, push from the top. If the load is too heavy for the machine, adjust the height or lower the bucket accordingly.



Handling Rubber Tracks

Due to the use of rubber, rubber tracks have inherent weaknesses, namely a lack of strength. Please strictly adhere to the following prohibitions and precautions to prevent track damage or detachment.

Prohibitions:

Do not walk or operate the machine in the following areas:

Walking and turning on crushed stones, extremely rough hard rocks, steel beams, scrap metal, or near the edges of steel plates may damage the rubber tracks.



Walking on riverbeds or areas with a large number of pebbles may cause stones to get stuck in the tracks and damage or detach the tracks.

Do not use the machine near the seaside as saltwater may corrode the steel core of the tracks.

Do not let fuel, lubricants, salt, or chemical solvents adhere to the tracks. These substances may corrode the welded areas of the steel core and cause rust or peeling. If any of these substances come into contact with the tracks, immediately clean them with water.

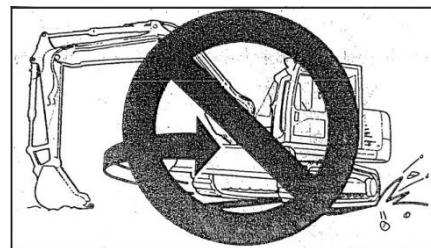
If the machine walks on irregular surfaces, is exposed to bonfires, or walks on hot iron plates under direct sunlight, it may cause irregular wear or damage to the tracks.

Do not move the rubber tracks on ground that may cause slipping, as this may accelerate track wear.

Safety precautions:

When operating this machine, please observe the following safety precautions:

Do not rotate the swing frame (turntable) while lifting the front of the machine with the bucket operation. Rotation in this manner may cause rapid damage to the rubber tracks due to the load being concentrated on a single point of the tracks.



Avoid sudden changes in direction or turning in

place on concrete surfaces as this may cause wear or damage to the rubber tracks.

Salt, potassium chloride, ammonium sulfate, potassium sulfate, and calcium water phosphate will damage the tracks. If any of these substances come into contact with the tracks, immediately clean them thoroughly with water.

Do not allow the sides of the rubber tracks to rub against concrete or walls.

Avoid hitting the rubber tracks with the bucket, as it may cause damage to the tracks.

Be cautious on snowy or frozen surfaces during winter, as the tracks may easily slip under such conditions.

Use the rubber tracks within the temperature range of -25°C to +55°C.

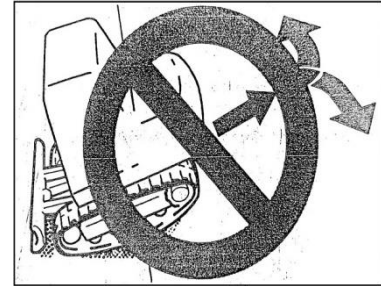
For long-term storage of rubber tracks (three months or longer), store them indoors and avoid exposure to direct sunlight or rain.

As the entire track is made of rubber, it is not as stable as steel tracks, so exercise special caution during lateral rotation and swinging.

Preventing rubber track detachment:

Adhere to the following precautions to prevent track detachment:

Always maintain the appropriate tension of the tracks. When the vehicle needs to cross larger steps, such as cobblestones or rocks (20 cm or deeper), approach the steps at a right angle and avoid changing directions on top of the steps.



During reverse climbing on slopes, avoid changing direction at the starting point of the slope.

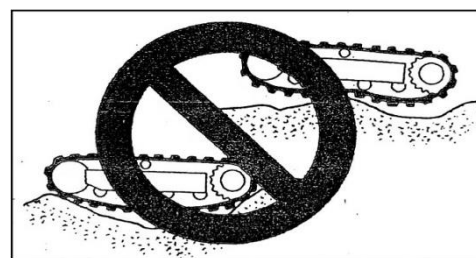
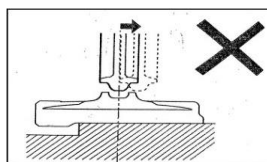
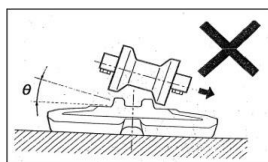
Avoid walking with one track on a slope or protrusion and the other track on a flat surface (causing the machine to tilt at an angle of 10° or more).



Instead, walk with both tracks on a flat surface. Do not change direction when the tracks are loose as shown in the diagram.

In this situation, if the machine moves backward, the rubber track may detach.

In this situation, if the machine rotates, the rubber track may detach.

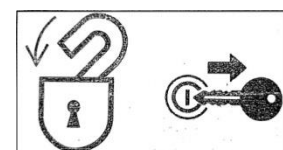


Precautions when stopping

Safe Parking

Park the machine on a level, solid, and secure ground, and engage the parking brake.

If parking on a slope or with the machine inclined, securely immobilize the machine to prevent it from moving.



When parking on the street, use barriers, warning signs, lights, etc., to make the machine visible even at night and avoid collisions with other vehicles.

Before leaving the machine, perform the following tasks: lower the bucket and blade to the ground. Turn off the engine and remove the ignition key. Lock the cabin and compartments and take the key with you.

Main structural components of the machine

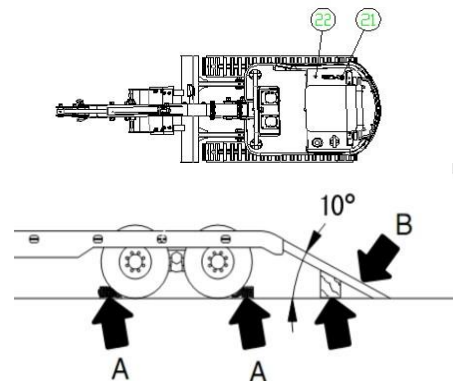
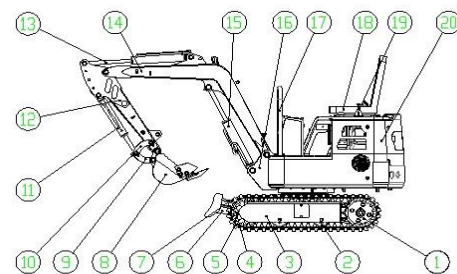
No.	Name	No.	Name	No.	Name	No.	Name
1	Drive Wheel	7	Dozer Blade Cylinder	13	Stick Cylinder	19	Power Switch
2	Rollers	8	Bucket	14	Boom	20	Cover
3	Lower frame	9	Rocker arm	15	Boom cylinder	21	Throttle
4	Track	10	Connecting rod	16	Upper frame	22	Key start switch
5	Idler wheels	11	Bucket cylinder	17	Armrest		
6	Dozer Blade	12	Stick	18	Seat		

Precautions when transporting

Safe Loading/Unloading of the Machine

During the loading and unloading process, the machine may overturn or fall. Please take the following safety measures:

1. Choose a solid and level ground and keep an adequate distance from the roadside.
2. Securely fix a ramp with sufficient strength and size to the truck bed. The slope of the ramp should not exceed 15°. If the ramp bends too steeply downward, support it with props or blocks.
3. Do not use the working equipment to load or unload the machine, as it may cause the



machine to tip over or fall.

4. Keep the truck bed and loading surface clean, free from oil, sand, ice, snow, or other foreign materials to prevent the machine from sliding. Clean the tracks.
5. Use chocks to prevent the transport vehicle from moving.
6. During the loading and unloading process, follow the signals from the signalman and proceed slowly in low gear.

Do not change the route on a slope.

Do not rotate or swing on a slope, as the machine may overturn.

When rotating or swinging on the truck bed, the base may not be stable enough, so proceed slowly.

If possible, lock the doors and covers after loading. Otherwise, they may open during transportation.

Secure the tracks with wedges and then fasten the machine to the truck bed with ropes or chains.

Safe lifting of the machine:

Master and use the correct lifting gestures.

Check the lifting equipment daily for any damaged or missing parts and replace them as necessary.

Use cables or ropes capable of lifting the weight of the machine during lifting.

Follow the lifting procedures as described in the manual; do not perform any other operations, as it may cause the machine to become unbalanced.

Do not lift the machine when there are operators on it.

Lift the machine slowly during the lifting process to prevent overturning.

Keep all personnel away from the working area during lifting and do not move the machine above people's heads.

Safe transportation of the machine:

When transporting the machine, you should be familiar with and comply with applicable safety regulations, vehicle codes, and traffic rules.

Consider the length, width, height, and weight of the truck carrying the loaded machine to choose the best transportation route.

Avoid sudden acceleration, sudden stops, or high-speed driving during transportation, as it may cause the loaded machine to shift or lose balance.

Maintenance Precautions

"DO NOT OPERATE" Warning Information

When inspecting or maintaining the machine, unauthorized starting of the engine or touching the control levers may cause serious accidents.

Before performing maintenance, turn off the engine, remove the key, and keep it with you.

Place "DO NOT OPERATE" warning labels prominently on the starting switch, control levers, and other visible locations.



Use the correct tools.

Do not use damaged or deteriorated tools or tools designed for other purposes. Use appropriate tools for the specific tasks.

Regular Replacement of Safety Critical Components

To ensure the machine can be used safely for an extended period, regular refueling, inspection, and maintenance are essential. For enhanced safety, please schedule regular replacements of safety critical components such as hoses and seat belts.

"Regularly replaced safety critical components" refer to parts that may become aged, worn, or degraded in functionality after repeated use. The performance of such components can change over time, which may lead to severe mechanical damage or personal injury. Judging the remaining lifespan of these components based solely on visual inspection or operational feel is challenging.

If any damage is observed during visual inspection, promptly replace the damaged parts, even if the designated replacement interval has not been reached.

Regularly replace fuel hoses. Fuel hoses can wear out over time, even if no visible signs of wear are present.

To ensure safe machine operation, perform regular inspections and maintenance. The following safety critical components must be regularly replaced to enhance safety.

Damage to these components can cause severe injuries or fires.

List of Safety Critical Components:

Machine Body	Safety critical components that require regular replacement.	Replacement Interval
Fuel System	Fuel Line	Every 2 years
	Filler Cap on Fuel Tank	

Hydraulic System	Main Unit	Hydraulic Hose (Pump Outlet)	
		Hydraulic Hose (Pump Inlet)	
		Hydraulic Hose (Swing Motor)	
		Hydraulic Hose (Travel Motor)	
	Working Equipment	Hydraulic Hose (Boom Cylinder)	
		Hydraulic Hose (Arm Cylinder)	
		Hydraulic Hose (Bucket Cylinder)	
		Hydraulic Hose (Blade Cylinder)	
		Hydraulic Hose (Auxiliary Circuit)	

Explosion-proof lighting

When inspecting fuel, lubricants, coolant, or battery electrolyte, please use explosion-proof lights to prevent fires or explosions that could lead to serious injuries or fatalities.

Unauthorized personnel are strictly prohibited from entering.

During operations, do not allow unauthorized personnel to enter the work area. Exercise caution when performing grinding, welding, or hammering tasks, as flying debris from the machine may cause injuries.



Worksite preparation:

Choose a stable and level work area with adequate lighting conditions. If working indoors, ensure proper ventilation.

Clear away any obstacles and hazardous materials, and eliminate slippery areas.

Always keep the machine clean.

Before maintenance, clean the machine.

Before cleaning the machine, turn off the engine and cover electrical components to prevent water ingress. Water entering electrical components may cause short circuits or malfunctions. Do not use water or steam to clean the battery, electronic control elements, sensors, connectors, or the operator's cab.

Before performing maintenance, turn off the engine.

Avoid lubricating or making mechanical adjustments to the machine while it is running or when the engine is running, even if the machine is not in operation.

If maintenance is necessary while the engine is running, arrange for two people to work as a team and maintain communication with each other.

One person must remain seated in the operator's seat so that they can immediately shut off the engine if necessary. This person should be particularly careful not to touch the control levers and pedals, except when required.

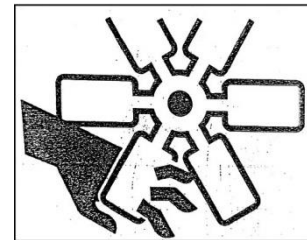
The other person performing maintenance must ensure that their body or clothing is kept away from moving parts of the machine.

Stay away from all rotating and moving parts.

Getting hands or tools caught in rotating or moving parts can lead to serious injuries or even fatalities.

Do not throw or insert any objects into the fan or fan belt, as they may be blown apart or cut. Do not throw or insert anything into the fan or fan belt. Securely fasten the machine or any parts that may fall off.

Before performing maintenance or repairs under the machine, lower all movable attachments to the ground or to their lowest position.

**Fixing the tracks**

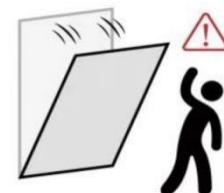
If it is necessary to work under a raised machine or equipment, it must be securely supported with wooden blocks, jacks, or other sturdy and stable supports. Do not go under the machine or work attachment until it is securely supported. This procedure is particularly important when performing hydraulic cylinder operations.

Stabilizing the work attachment

When performing maintenance or replacing bucket teeth or side teeth, the work attachment should be firmly secured to prevent the machine from moving accidentally.

Secure the engine hood or cover when opening it.

Before performing any work inside the machine, make sure to secure the engine hood or cover. Keep the engine hood or cover closed in windy conditions or when the machine is parked on a slope.



Place heavy objects in a stable position

During disassembly or installation, when heavy objects or attachments need to be temporarily placed on the ground, be sure to place them in a stable location and prevent unauthorized personnel from approaching the area where such items are stored.



Refueling Notes

Do not smoke or have open flames while refueling or near the refueling area.

Do not remove the fuel tank cap or refuel when the engine is running or has not cooled down, and avoid spilling fuel onto hot surfaces of the machine.

Refuel the fuel tank in a well-ventilated area.

Do not fill the fuel tank to the brim; leave room for fuel expansion.



Clean up any spilled fuel immediately.

Tighten the fuel tank cap securely. If the fuel tank cap is lost, only replace it with an original one. Using a non-authorized and poorly-ventilated fuel tank cap can cause internal pressure in the fuel tank.

Prevent dust from entering

When installing and removing components, it should be done in a dust-free area. Clean the work area and the components to prevent dust from entering.

Clean the mounting surface

When installing and removing components, ensure that the contact surfaces of the components are clean. If the sealing groove on the contact surface is damaged, please contact your sales or service agent for repair or resolution.

Seals and Cotter Pins

Always replace all removed seals and cotter pins with new ones.

During installation, be careful not to damage or twist the seals.

Seals

When wrapping the plug with seal tape, clean the old seal tape from the threads and wash the threads.

Tighten the seal tape around the threads, leaving 1 or 2 turns uncoiled at the end of the plug.

Do not use fuel for cleaning.

Use the correct fuel grade and lubricants according to the season.

Refer to the table below and choose the appropriate fuel, lubricants, and grease based on the temperature. Regardless of the designated time, if the oil becomes too dirty or has deteriorated, please replace it.

When refueling, do not mix different brands of oil. If you want to change the brand, replace all fuel/lubricants.

Fuel

Diesel fuel should meet the following specifications. The table lists several current diesel fuel specifications worldwide.

Diesel specification		Diesel specification	
GB252	China	BS2869-A1 or A2	U.K.
ASTMD975 Serial number: 1-D,S15	USA Canada	ISO8217DMX	International
Bio-diesel Bio-diesel blended as B5ASTMD6751, D7467.			
EN590:96	European Union	JISK22042 Level	Japan
Bio-diesel Bio-diesel blended as B5ENI4214,EN590			

To maintain engine performance and longevity, always use clean, high-quality fuel.

To prevent freezing in cold weather, select diesel that remains usable at least 12°C lower than the expected lowest outdoor temperature.

Use diesel with a cetane number of 45 or higher, and in extremely cold or high-altitude areas, use diesel with even higher cetane values.

Use fuel with a sulfur content of volume ratio less than 0.05-0.0015%. (In the USA or Canada, use ultra-low sulfur fuel). Using high-sulfur fuel may cause sulfuric acid corrosion in the engine cylinder.

Prohibit the use of kerosene. Do not mix kerosene, used engine lubricating oil, or residual fuel with diesel.

Poor-quality fuel can reduce engine performance or cause engine damage. The use of fuel additives is not recommended, as some additives may result in decreased engine performance.

Metal content, such as zinc, sodium, magnesium, silicon, and aluminum, should be limited to one part per million by mass (1 mass ppm) or lower.

When using biodiesel, the warranty provisions of engine manufacturers become

invalid for machines using biodiesel that does not meet the standards or has degraded.

Lubricating oil

API standard: American Petroleum Institute

ACEA standard: European Automobile Manufacturers Association

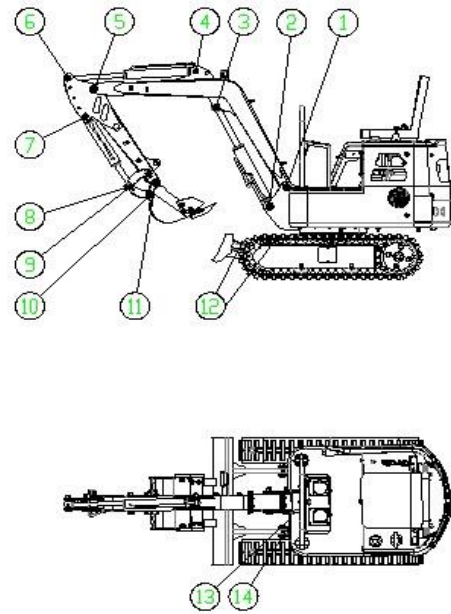
SAE standard: Society of Automotive Engineers

Part	Type	Types of lubricants selected based on temperature -41432506886104°F -20-10010203040°C	Replacement time interval.
		ISOVG32 ISOVG46 ISOVE68 50% coolant mixture 30% coolant mixture	
Travel reduction gear	Gear Oil API:GL-4	SAE90	Every 1000 hours after the first 250 hours.
Rotary motor gear	Lithium-based grease EP-2 NLGI2#	-	Every 50 hours
Slewing ring			Every 50 hours
Working equipment			Daily or every 10 hours
Rod			As needed

1. If the walking time accounts for a high proportion of the total operating time, the gear oil should be replaced before the specified time.
2. Note: When using water, please use tap water (soft water), do not use well water or river water. When the outdoor temperature drops below 0°C, add coolant (antifreeze). Determine the mixing ratio according to the coolant manufacturer's instructions.
3. Note: The hydraulic oil change interval depends on the type of hydraulic oil being used. When using regular anti-wear hydraulic oil, the hydraulic oil should be changed every 2000 hours.

Lubrication diagram for key machine structures

No.	Lubrication Points
1	Boom, upper frame pivot pins
2	Boom cylinder, upper frame pivot pins
3	Boom, boom cylinder pivot pins
4	Boom, arm cylinder pivot pins
5	Boom, arm pivot pins
6	Arm, arm cylinder pivot pins
7	Arm, bucket cylinder pivot pins
8	Link, bucket cylinder pivot pins
9	Arm, link pivot pins
10	Bucket, boom pivot pins
11	Arm, bucket pivot pins
12	Blade cylinder pivot pins
13	Swing bearing outer teeth grease ports
14	Swing bearing inner teeth grease ports



Handling of hoses:

1. Lubricant or fuel leakage can cause a fire. Take necessary precautions to prevent leakage and ensure proper maintenance of hoses and connections.
2. Do not twist, bend, or impact hoses. Avoid any physical damage to the hoses as it can compromise their integrity and lead to failure.
3. Do not use twisted, bent, or cracked pipes, metal tubes, or hoses, as they may rupture or fail under pressure. Replace damaged hoses promptly.
4. Re-tighten loose fittings and connections to prevent leaks and ensure a secure and tight fit.
5. Regularly inspect hoses for signs of wear, damage, or degradation. Replace worn-out or damaged hoses to maintain safety and optimal performance.

Be extremely careful when handling high-temperature and high-pressure components:

1. Always turn off the engine and allow the machine to cool down before performing any maintenance. Engine components, exhaust pipes, radiators, hydraulic hoses, sliding parts, and many other parts of the machine can be very hot just after the engine is turned off, and touching these parts can cause burns.



2. Take caution with engine coolant, hydraulic oil, and other fluids as they are also under high temperature and pressure. Avoid touching hydraulic oil when loosening plugs or connections, as hot oil can spray out and cause burns or injury.



3. When working with the cooling system, be careful as the coolant can be scalding hot. Do not remove the radiator cap or drain plug while the coolant is hot. Turn off the engine and wait until both the engine and coolant have cooled down before slowly loosening the radiator cap to release internal pressure before removing it.

Be cautious of hydraulic oil pressure:

After the engine is turned off, hydraulic oil pressure in the hoses can remain for a long time. Before conducting maintenance operations, it is essential to thoroughly release the internal pressure.



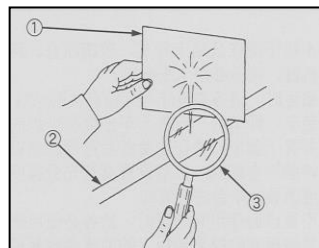
Hydraulic oil under high pressure can penetrate the skin or eyes, causing severe injuries, blindness, or even death. Remember that hydraulic oil leaking from small holes is almost invisible to the naked eye. When checking for leaks, always wear protective goggles and thick gloves and use cardboard or plywood to protect your skin to avoid injury from hydraulic oil spray.

If hydraulic oil enters the skin, it must be removed within a few hours by a doctor experienced in treating such injuries using surgical methods.

Before performing any work on the hydraulic system, it is essential to first release the pressure.

Before releasing the pressure in the hydraulic system, such as removing covers or filters or disconnecting pipes, hydraulic oil may spray out.

Slowly loosen the vent plug to release the pressure in the oil tank.



When removing plugs, screws, or disconnecting hoses, stand to one side and gradually loosen them to gradually release the internal pressure before fully removing them.

Engine oil or oil plugs may spray out due to pressure in the travel motor tank. Please slowly loosen the oil plug to release the internal pressure.

Be cautious of flying debris when using a hammer.

When using a hammer, nails or metal fragments may fly around, which can cause serious injuries.

When using a hammer to strike nails, bucket teeth, side teeth, or other hard metal parts such as bearings, please wear protective gear such as safety goggles and gloves. Ensure that there are no people nearby when striking nails or bucket teeth.

Be cautious of high-pressure grease.

In the track tensioning device, when adjusting the tensioning force under high pressure, failure to follow the specified procedure may cause the grease relief valve to fly out, resulting in injuries.



Do not loosen the grease fittings.

Slowly release the grease relief valve and do not turn it more than one turn.

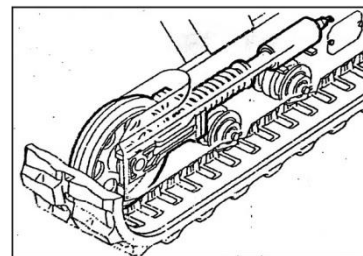
Do not place your face, arms, legs, or body in front of the grease relief valve.

If no grease flows out when loosening the grease relief valve, the valve may be faulty.

Please contact sykend service agent for repairs.

Do not disassemble the track tensioning device.

The track tensioning device is equipped with a powerful spring. Accidentally disassembling the track tensioning device may cause the spring to bounce out, leading to serious injuries.



Disconnect the battery cable.

Before performing electronic system work or welding, please disconnect the battery cable. First, disconnect the negative (-) battery cable, and when reconnecting, connect the negative (-) battery cable last.

Handle the battery with care.

The battery contains sulfuric acid, and accidental contact can cause harm to the eyes or skin.

If it comes into contact with the eyes, immediately flush with water and seek medical attention promptly.

If swallowed accidentally, drink plenty of water or milk and seek immediate medical attention.

If sulfuric acid comes into contact with the skin or clothing, immediately rinse with plenty of water.

When handling the battery, wear safety goggles and gloves.

The battery can generate flammable hydrogen gas, which may cause an explosion.

Keep away from open flames, sparks, or lit cigarettes.

Use a flashlight when checking the electrolyte level.

Before checking or handling the battery, ensure to turn off the ignition switch and stop the engine.

Be careful not to allow metal tools or any metal objects to come into contact with the battery terminals, as it may cause a short circuit.

Tighten the terminals securely to prevent sparks from loose connections.

Ensure the battery cover is tightly sealed.

If the battery is frozen, do not charge it or attempt to start the engine, as it may cause an explosion. Warm the frozen battery to 15°C before use.

Do not use the battery when the electrolyte level is below the lower limit, as it can accelerate internal aging and shorten battery life, potentially leading to rupture (explosion).

Do not fill the battery with distilled water above the upper limit, as it can cause the electrolyte to leak and corrode machine parts.

Use a damp cloth to clean the area around the electrolyte level line and check the level. Do not use a dry cloth, as it may cause static electricity buildup, leading to combustion or explosion.

Using the battery charging line to start the engine:

When using the battery charging line to start the engine, follow the correct steps for connecting the cables to avoid discharges and battery explosions due to incorrect connections.

Avoid letting the "problem machine" and "rescue machine" touch each other.

Do not let the positive (+) and negative (-) ends of the battery charging line touch or come into contact with the machine.

When connecting, first connect the positive end of the battery charging line to the



positive (+) terminal. When disconnecting, first disconnect the negative end of the cable from the negative (-) terminal (ground terminal). Ensure that the cable connections are secure. Connect the last cable clamp of the battery charging line as far away from the battery as possible.

Always wear safety goggles and gloves when using the battery charging line to start the engine.

Use battery charging lines and clamps that are the appropriate size for the battery capacity. Do not use damaged or corroded battery charging lines and clamps.

Ensure that the battery capacity of the "rescue machine" is the same as that of the "problem machine."

Please entrust the welding repair to a qualified sykend service agent.

When welding is necessary, it must be carried out by qualified personnel in a well-equipped workplace. To prevent damage to any parts of the machine caused by excessive current or sparks, please follow these guidelines. Disconnect the battery wiring before performing welding. Do not apply a continuous voltage of 200V or higher. The grounding point must be connected within 1 meter of the welding area. Do not connect the grounding point near electronic control devices/instruments or connectors. Ensure that there are no seals or bearings between the welding area and the grounding point.

Do not connect the grounding point around the pins of the work equipment or hydraulic cylinders.

When performing welding work on the machine body, disconnect the connectors of the electronic control device before starting the operation.

Vibration exposure to the operator:

Test results on the vibration transmitted to the operator from the machine show that the vibration experienced by the operator's upper limbs is below 2.5m/s^2 , and at the seat, the vibration experienced by the body is below 0.5m/s^2 .

Waste disposal

Ensure that the waste oil discharged from the machine is collected in a container. Improper disposal of waste oil can be harmful to the environment. When handling hazardous substances such as lubricating oil, fuel, coolant, solvents, filters, and batteries, please comply with applicable laws and regulations.



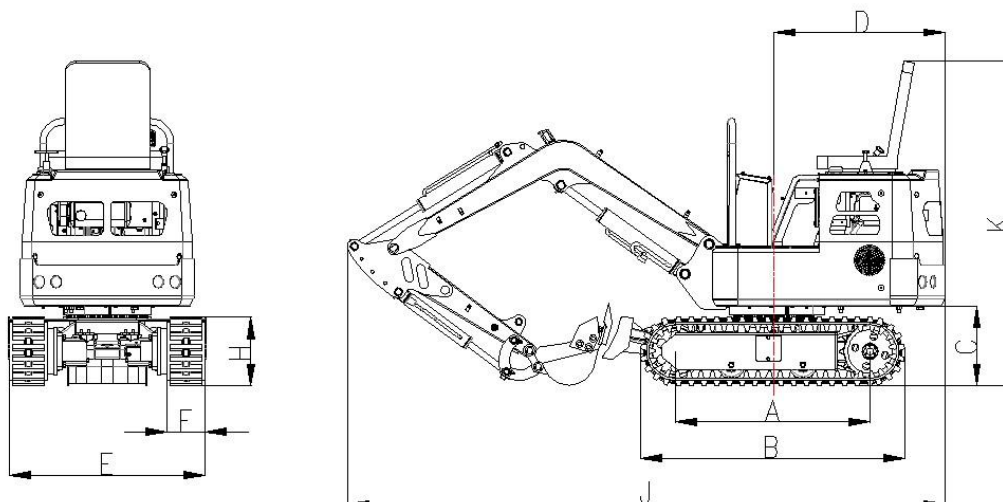
Handling of hazardous chemicals

Direct contact with hazardous chemicals can cause serious injuries. The hazardous chemicals used in this machine include grease, battery electrolyte, coolant, paints, and adhesives. Please handle hazardous chemicals with caution and care.

Technical specifications for HW10 series excavators

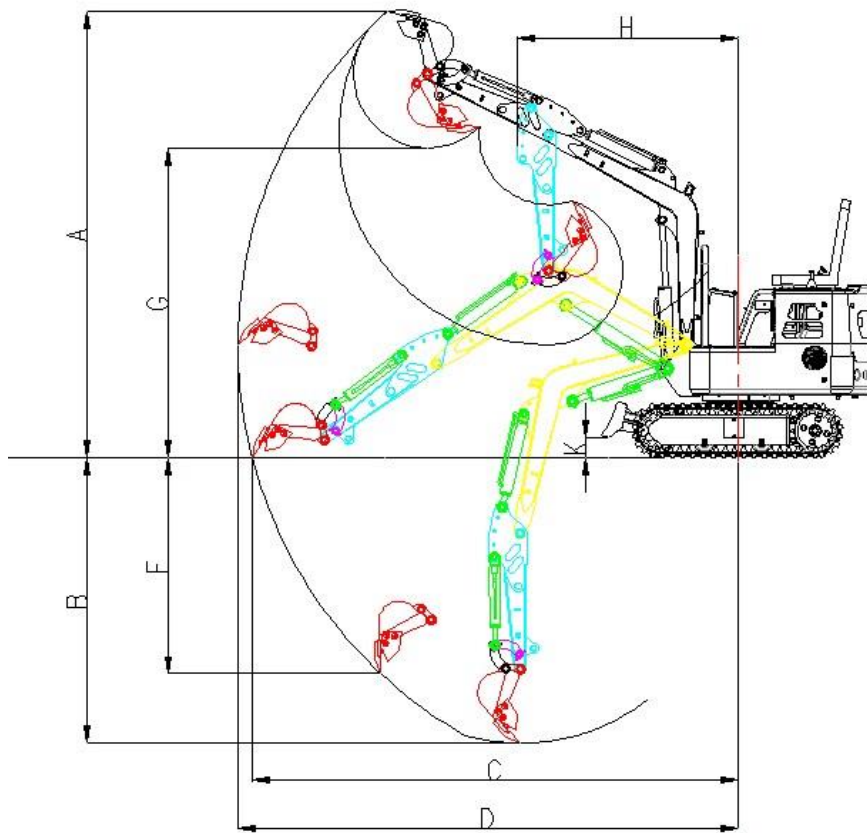
	Technical parameter		Unit	Value
Total Weight			kg	850
Bucket capacity			m ³	0.025
Work device type				Backhoe
Engine	Model			192FAM
	Displacement		L	0.499
	Rated output power / speed		kW/r/min	8.6/4000
	Maximum torque		N.m/r/min	25/2880
Speed	Maximum walking speed		km/h	2
	Swing speed		rpm	13
	Maximum gradeability			25°
	Digging force		kN	6.5
Size	Total length		mm	2770
	Total width		mm	930
	Total height		mm	2080
	Dozer Blade	Maximum Clearing Depth	mm	270
		Maximum ground clearance height	mm	140
Track parts	Ground specific pressure		Kgf/cm ²	0.35
	Track	Material		Rubber
		Tension control type		Butter tank
Pump type				Gear pump
Hydraulic fluid capacity	Work pressure		Mpa	16
	Flow		(L/min)	21.6
	Hydraulic oil tank capacity		(L)	18
	Fuel tank capacity		(L)	6.5
	Engine oil capacity		(L)	1.65

Main specifications of HW10 series excavators.



Machine size		Unit: mm
A	Wheelbase	915
B	Track overall length	1240
C	Ground clearance of platform	370
D	Tail swing radius	800
E	Chassis width	930
F	Track width	180
H	Track height	320
J	Transport length	2770
K	Overall height	1520

The working range of the HW10 series excavator

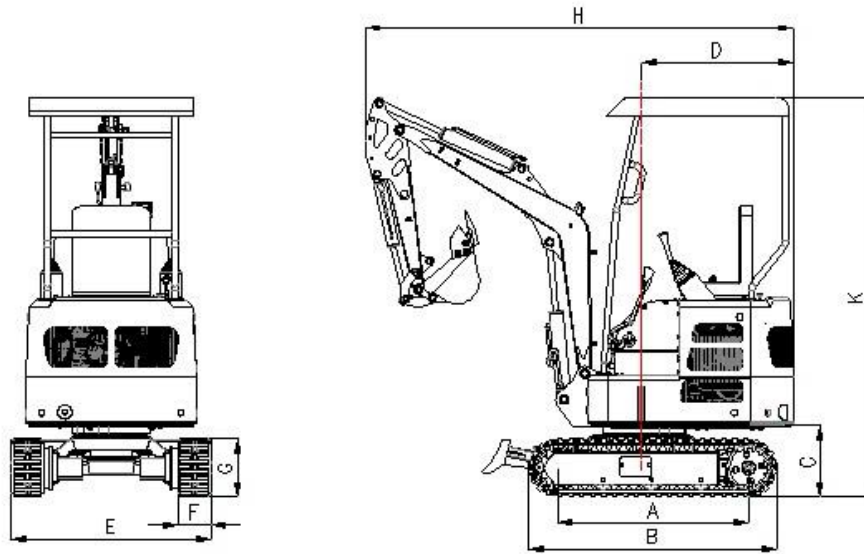


Working range		Unit: mm
A	Maximum digging height	2620
B	Maximum digging depth	1675
C	Maximum ground digging radius	2960
D	Maximum digging radius	2945
F	Maximum vertical digging depth	1268
G	Maximum dumping height	1900
H	Minimum swing radius	1300
K	Maximum lifting height of the bulldozer blade	140

Technical specifications of WY17 series excavator

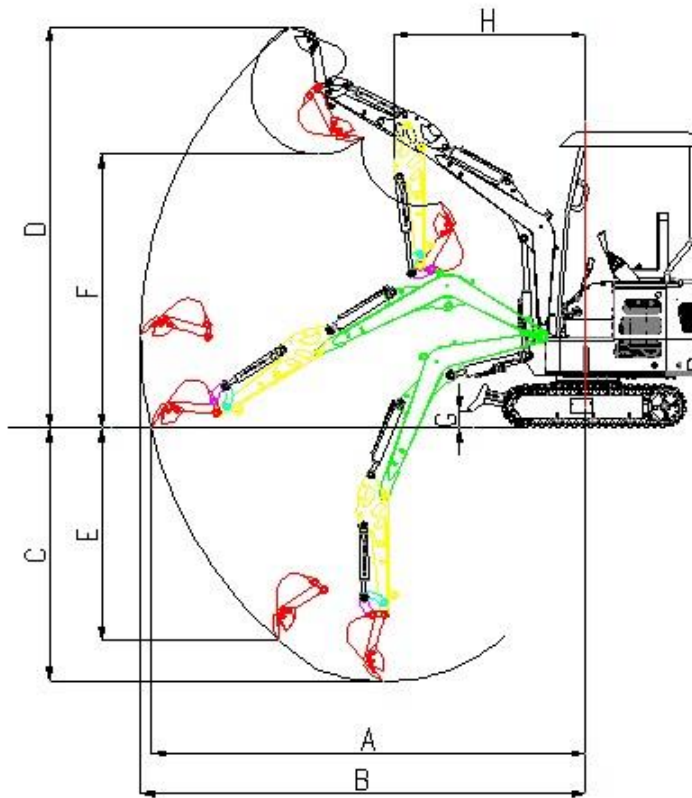
	Technical parameter		Unit	Value
Whole machine weight			kg	1110
Bucket capacity			m ³	0.04
Working device type				Backhoe
Engine	Model			KM385B-216A1
	Displacement		L	1.532
	Rated power / speed		kW/rpm	26/3300
	Maximum torque / speed		N·m/rpm	85/2110~2310
Speed	Maximum walking speed		km/h	2.5
	Swing speed		rpm	8-10
	Maximum gradeability		°	30
	Digging force		kN	10.5
Dimensions	Total length		mm	3000
	Total width		mm	1100
	Total height		mm	2270
	Dozer Blade	Maximum clearing depth	mm	190
		Maximum ground clearance height	mm	140
Crawler components	Ground specific pressure		kpa	29.8
	Track	Material		Rubber
		Tension control type		Butter tank
Pump type				Gear pump
Hydraulic fluid capacities	Work pressure		Mpa	16
	Flow		(L/min)	30
	Hydraulic oil tank capacity		(L)	22
	Fuel tank capacity		(L)	10.5
	Engine oil capacity		(L)	3.8

Main specifications of the WY17 series excavator



Machine size		Unit: mm
A	Wheelbase	1050
B	Track overall length	1365
C	Ground clearance of platform	390
D	Tail swing radius	835
E	Chassis width	1100
F	Track width	180
G	Track height	320
H	Transport length	2300
K	Overall height	2270

The working range of the HW10 series excavator



Working range		Unit: mm
A	Maximum ground excavation radius	3220
B	Maximum excavation radius	3300
C	Maximum excavation depth	1883
D	Maximum excavation height	2975
E	Maximum vertical digging depth	1577
F	Maximum unloading height	2028
G	Maximum lifting height of the bulldozer blade	126
H	Minimum turning radius	1425

Diagnosis and Troubleshooting of Common Faults

In order to ensure the excellent quality of sykend excavator equipment, we use high-quality components for all parts. The performance and service life of the equipment depend not only on the manufacturing and assembly quality but also on the maintenance and care of the equipment.

Our excavator sales representatives and service engineers have an obligation to remind users that preventive maintenance and care are the easiest and most economical forms of maintenance.

The maintenance and care of the excavator are divided into daily inspections and maintenance as well as major, medium, and minor periodic maintenance, based on their frequency.

Diagnosis and Troubleshooting of Common Mechanical System Faults

Fault Symptoms	Cause analysis	Troubleshooting methods
Excessive noise from structural components	Loose fasteners causing abnormal noise. Increased wear in the gap between the bucket and the boom end face.	Check and tighten to the end. Adjust the clearance to be less than 1mm.
Bucket teeth falling off during operation	Insufficient elasticity of the used bucket tooth pins, causing multiple replacements. Mismatch between the bucket tooth pin and the tooth holder.	Replace the bucket tooth pin.
Tracks getting tangled underneath the excavator	Slack tracks. Front wheels driving at high speeds on rough roads.	Load the tracks. Drive the front idler slowly on rough roads.

Diagnosis and Troubleshooting of Common Hydraulic System Faults

Fault Symptoms	Cause analysis	Troubleshooting methods
Excavator has no movement at all	The hydraulic oil level in the tank is insufficient, causing the main pump to suck in air.	Add enough hydraulic oil.
	The suction filter is clogged.	Replace the filter and clean the system.
	The engine coupling is damaged.	Replace

	The main pump is damaged.	Replace or repair the main pump.
	The relief valve is set to a too low pressure or is stuck.	Adjust to the normal pressure or disassemble for cleaning.
	The main pump suction hose is burst or disconnected.	Replace with new pipes.
One side track cannot move	The main valve supplying oil for the movement of one side of the tracks is damaged.	Replace
	The main valve spool is stuck or the spring is broken.	Repair or replace
	The travel motor is damaged.	Replace
	There is communication between the upper and lower chambers of the rotary joint.	Replace the oil seal
	The travel oil hose is burst.	Replace
Excavator has weak overall movements	The hydraulic oil level in the tank is insufficient.	Fill with hydraulic oil
	The engine's RPM is too low.	Adjust the engine speed.
	The system's safety valve pressure is too low.	Adjust to the specified pressure.
	The main valve has severe internal leakage.	Replace the main pump or repair it.
	The travel motor, rotary motor, and cylinders all have varying degrees of wear, leading to internal leakage.	Replace or repair worn parts.
	The engine's fuel filter is clogged, causing significant speed reduction and potential engine stalling in severe cases.	Replace the filter element.
	The hydraulic oil filter is clogged, accelerating wear on the pump, motor, and main valve, leading to internal leakage.	Regularly replace the filter element according to maintenance requirements.

	The main valve and valve hole clearance are excessively worn, resulting in severe internal leakage.	Repair the valve stem.
No movement in either direction of travel (other functions work normally)	The central rotary joint is damaged.	Replace the oil seal
	The travel control valve has severe internal leakage.	Replace
	The pressure of the main valve's travel relief valve is too low or the valve spool is stuck.	Adjust and grind
	The left or right travel reducer is malfunctioning.	Repair
	The left or right travel motor is malfunctioning.	Repair
	Oil hose burst.	Replace
The machine veers off course while moving (other functions work normally)	There is an error in adjusting the variable point of the main pump or there is excessive internal leakage in one of the pumps.	Adjust or repair
	There is damage or jamming in the inner or outer spring of one of the travel valve cores in the main valve.	Replace
	The travel motor has wear and is producing internal leakage.	Repair or replace
	The central rotary joint's sealing components are aging and damaged.	Replace the seals
	The left and right tracks have different levels of tightness.	Adjust
Boom (arm, bucket) only moves in one direction	The main valve spool is stuck or the valve stem spring is broken.	Repair or replace
Boom (arm, bucket) has no movement	The main valve's arm valve stem is stuck or the overload pressure is too low.	Repair
	There are oil leaks, disconnections, damaged O-rings, or loose pipe joints in the supply line.	Replace the damaged parts

	There are sandholes inside the main valve, causing communication between high and low-pressure chambers.	Replace
Boom (arm, bucket) drops too fast or when not operated at a certain height, the hydraulic cylinder falls under its own weight	The overload valve pressure is too low.	Adjust
	The oil cylinder has significant internal leakage.	Replace the seals, repair the scratches and grooves on the inside wall of the hydraulic cylinder, or replace the hydraulic cylinder.
	The oil pipe joints are loose, and the O-rings are damaged.	Replace
The boom (arm, bucket) works slowly and lacks power	The multi-way valve has severe internal leakage or sandholes.	Replace
	The overload pressure is low.	Adjust
	The oil cylinder has significant internal leakage.	Replace the oil seal
	The main pump has internal leakage and is not functioning properly.	Repair or replace
There is movement in the boom (arm, bucket) when not operated	The multi-way valve spool is jammed or has excessive internal leakage.	Grind or replace
	The multi-way valve stem spring is broken.	Replace
	The working oil cylinder is leaking, causing the equipment to descend under its own weight.	Replace the oil seal
	The main valve overload relief valve pressure is too low or the spring is broken.	Adjust to the specified pressure. If the spring pressure is broken, it should be replaced.
Hydraulic oil temperature is too high	The correct grade of hydraulic oil recommended for the excavator is not used.	Replace the hydraulic oil

Hydraulic oil temperature is too high	The hydraulic oil cooler's external surface is contaminated with oil and excessive dirt, and the ventilation holes are blocked.	Clean the system
	The hydraulic oil level in the oil tank is too low.	Fill up the hydraulic oil
	The radiator's cooling function has failed.	Replace
No action during rotation (other functions work normally)	Hydraulic oil pipe is ruptured.	Replace
	The main valve's rotation valve stem is jammed.	Repair
	The swing motor is damaged.	Replace
	The swing bearing is damaged.	Replace
Rotation is sluggish and lacks power (other functions work normally)	There is severe external leakage in the hydraulic oil pipe.	Replace
	The overload pressure of the multi-way valve for rotation is low.	Adjust
	The swing motor has severe internal leakage.	Repair or replace
Unusual noises and abnormal vibrations occur while the excavator is in operation	The hydraulic oil level in the tank is insufficient.	Replenish oil
	The oil contains water and excessive air.	Replace
	The safety valve of the multi-way valve is making noise.	Adjust
	The coupling is damaged.	Replace
	The filter is clogged.	Replace
	The suction pipe is drawing air.	Exhaust air
	The engine's speed is uneven.	Adjust
	The working device bearings are not lubricated or damaged.	Add lubrication sleeve or replace axial sleeve.

Diagnosis and Troubleshooting of Common Electrical Control System Failures

Engine does not start
Engine stalls during operation
Engine does not shut off
Automatic idle does not work
All working devices and travel cannot move

Engine does not start

Accident details		Engine does not start		
Oil pump system does not supply oil or supplies insufficient oil.	Engine speed is too low		Adjust to normal engine speed.	
	The main pump has a malfunction		Replace the faulty component.	
	The fuel tank has insufficient oil		Refill the fuel tank	
	Oil pipes are broken, oil pipe joints are loose, or O-rings are damaged.		Replace the damaged O-ring or tighten the loose oil pipe joint.	
Speculated causes and standard values under normal conditions.	Causes		Standard values under normal conditions, and reference values for fault diagnosis.	
	1	Insufficient battery capacity	Battery Voltage	State of Charge Indicated by Hydrometer Color
			12V	Green (If white, the battery needs to be replaced)
	2	Fuse F1, F11 malfunction	When a fuse is blown, a ground fault in the circuit is highly likely to occur. If the monitoring lamp on the control panel is not illuminated, check the power circuit between the battery and specific fuses.	
	3	Engine ignition switch malfunction	Turn the engine start switch to OFF position for preparation and keep it in the OFF position during the fault diagnosis.	
			Ignition Switch	Location Resistance
			Between 30 and 17	OFF 1MΩ Start up Below 1Ω

4	Starting relay E3 malfunction	Turn the engine start switch to OFF position for preparation and keep it in the OFF position during the fault diagnosis.		
		Pin	Resistance	
		Between 85 and 85	200-400Ω	
		Between 87 and 30	1MΩ or more	
		Between 87a and 30	Below 1Ω	
5	Safety lock switch malfunction (internally disconnected)	Turn the engine start switch to the OFF position for preparation and keep it in the OFF position during the fault diagnosis.		
			Lock Lever	Resistance Value
		Between 105 and ground	Loose	1MΩ
			Locking	Below 1Ω
6	Starter motor malfunction (internally disconnected or short-circuited)	Turn the engine start switch to the OFF position for preparation, and keep the engine running during the fault diagnosis. If all power, grounding, generated signals, and engine start inputs are normal, but only the engine start output is abnormal, then the engine start motor relay is faulty.		
		Engine starter motor	Engine start switch	Voltage
		Power supply: B terminal and ground	Start	20~30V
		Engine start input: C terminal and ground		20~30V
7	Alternator (AC generator) malfunction	Turn off the engine start switch for preparation and keep it in the ON position during the fault diagnosis.		
			Voltage	
			Below 1V	
8	Wire line speed break (detached or poor contact with the connector)	Turn off the engine start switch for preparation and keep it in the OFF position during the fault diagnosis.		
			Resistance	Below 1MΩ

	9	Wire line speed break (contact with the ground wire line)	Turn the engine start switch to OFF to prepare for the diagnosis, and keep it in the OFF position during the troubleshooting period.		
				Resistance	Below 1MΩ
	10	Wire line speed short circuit (contact with the 24V circuit)	Turn the engine start switch to OFF to prepare for the diagnosis, and keep it in the ON position during the fault period.		
			Voltage	Below 1V	

Engine stalls during operation

Accident details			Engine stalls during operation		
	Reason		Standard values under normal conditions and reference values for fault diagnosis		
	1	Wire disconnection (poor contact with the connector)	Turn off the engine start switch for preparation and keep it in the OFF position during the fault diagnosis.		
			Between CN-12T and CN-132F wire speed	Resistance value	Below 1MΩ
		Wire grounding fault (contact with the ground circuit)	Turn off the engine start switch for preparation and keep it in the OFF position during the fault diagnosis.		
			Between CN-12T and CN-132F wire speed.	Resistance value	Below 1MΩ

Diagnosis of Common Diesel Engine Faults

Diagnosis of Engine Not Starting Fault

Symptom:

When trying to start the engine, the starter motor turns, but the engine does not start.

Possible Causes:

1. Low battery voltage and insufficient charging.
2. Corroded or loose battery terminals.
3. Corroded or loose battery ground cable or poor engine grounding.

4. Faulty starter relay (starter solenoid).
5. Faulty ignition switch or starter motor.

Troubleshooting Methods:

1. Low battery voltage is likely due to not turning off all electrical devices in the vehicle after parking. Always make sure to turn off all electrical devices before leaving the vehicle. During driving, pay attention to charging the battery so that it remains fully charged when parked. If the low battery voltage prevents starting, you can replace the battery with a new one or parallel another battery to start the engine.
2. Clean the battery terminals and tighten the power cable clamps to ensure a reliable connection between the power cable and battery terminals.
3. Clean the ground terminal of the battery and ensure a reliable grounding. Make sure the engine is properly grounded.
4. Repair or replace the starter relay (starter solenoid).
5. Check and repair the ignition switch.
6. Check and repair the starter motor.
7. If the battery has been used for a long time and its internal resistance has increased, it may need maintenance or replacement. Regularly maintain and correctly charge the battery. If necessary, replace it with a new battery and ensure it is fully charged to facilitate smooth engine starting.

Diagnosing the failure of diesel engine no start due to lack of fuel

Phenomenon:

When starting the engine, the starter turns at a normal speed, driving the engine to rotate, but the engine cannot start.

Causes:

Lack of fuel in the fuel tank;

Malfunction in the fuel supply system;

Air, water, or foreign matter in the fuel system causing fuel delivery obstruction;

Malfunctioning fuel pump;

Engine failure.

Troubleshooting:

Add standard fuel to the fuel tank and start the engine to deliver fuel to the carburetor;
Inspect the fuel supply system pipelines, fuel filter, fuel pump, etc., and replace any blocked or damaged components to ensure smooth fuel supply;
Eliminate air from the fuel system; if air obstruction prevents the engine from starting, cool down the system if necessary;
Inspect the fuel pump to ensure proper fuel supply; with a well-functioning fuel pump providing sufficient fuel, the diesel engine is less likely to experience fuel supply failure or any three obstruction issues (air, water, and foreign matter).

Diagnosis of Engine Hard Starting Fault

Phenomenon:

The starter motor rotates normally, driving the engine to turn, but the engine is difficult to start;
The engine is difficult to start when cold;
The engine is difficult to start when hot.

Causes:

Dirty and clogged fuel filter;
Faulty fuel pump;
Incorrect fuel injection timing;
Low engine oil temperature, low intake air temperature;
Clogged air filter;
Fuel leakage in the fuel line;
Faulty starter motor;
Incorrect starting procedure;
Incorrect fuel grade;
Engine malfunction.

Troubleshooting:

Inspect and replace the fuel filter;
Inspect and replace the air filter element;
Inspect and adjust the fuel pump;
Check the fuel line and ensure unobstructed fuel supply;
Inspect the starter motor and starting control system for reliable operation;
Follow the correct engine starting procedure;

Use the appropriate fuel grade; if necessary, drain any water from the bottom of the fuel tank;
Conduct a thorough inspection and maintenance of the engine.

Diagnosing Starter Failure

Phenomenon:

When turning on the ignition switch, the starter does not turn;
The starter drive gear does not engage;
The starter drive gear does not disengage;
The engine starting speed is insufficient, and it rotates unevenly.

Causes:

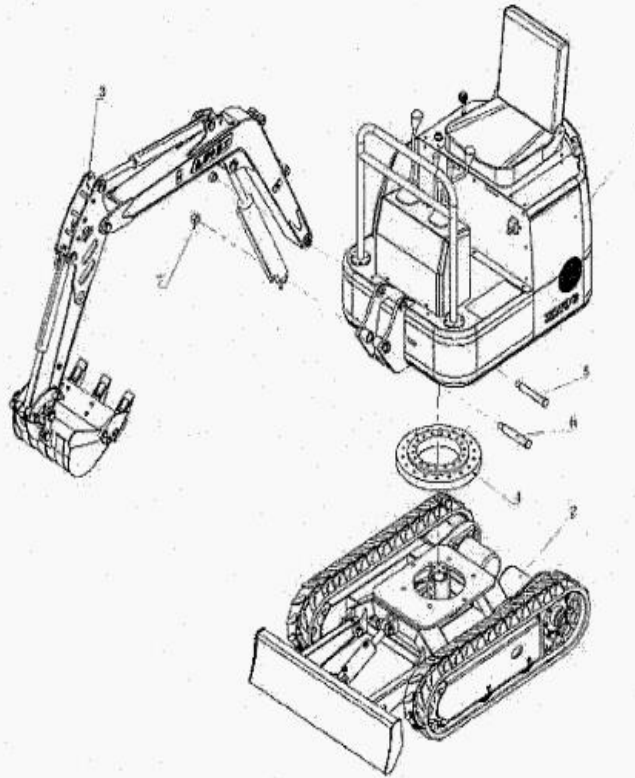
Insufficient charging of the battery;
Loose battery terminals;
Loose battery ground wire;
The starting circuit is not connected;
The electromagnetic relay armature is sticking;
Starter itself is faulty;
The starter drive gear and the engine flywheel ring gear are jammed;
The starter drive gear and bearings are sticking;
The starter cannot turn the engine;
Engine malfunction.

Troubleshooting:

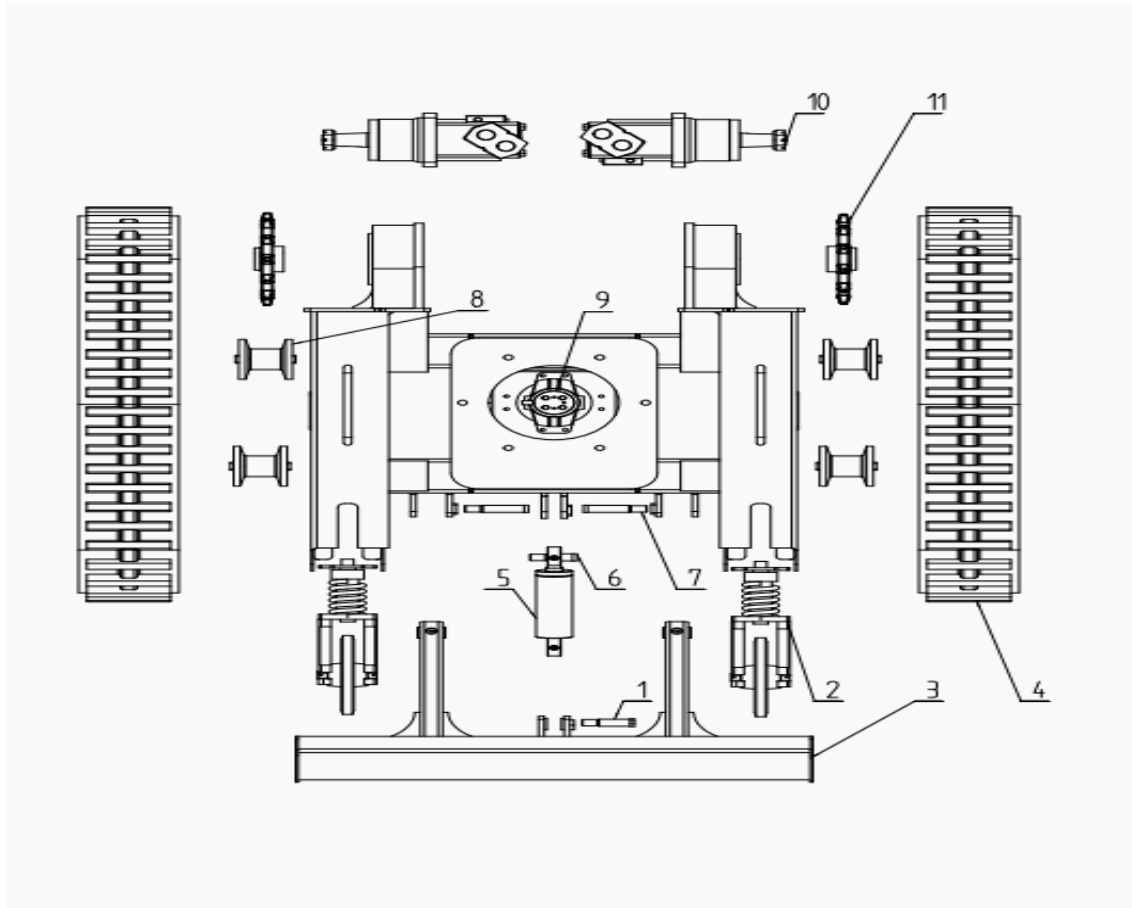
Check if the battery is sufficiently charged. If it is not, remove it and charge it, and replace the battery if necessary;
Ensure proper connection of battery terminals and connectors;
Repair the battery ground wire;
Check the starting circuit to ensure there is power at the starter terminal;
Inspect the starter electromagnetic relay and rectify any faults. When starting, there should be a distinct sound of the electromagnetic relay engaging and disengaging;
Inspect and repair the starter;
Reposition the starter drive gear and the engine flywheel ring gear to engage properly;
Overhaul the starter starting shaft end bearing;
If the starter torque is insufficient, consider replacing it;
Check and repair the engine to ensure it operates under normal conditions.

Bill of Materials (BOM) for the Entire Vehicle

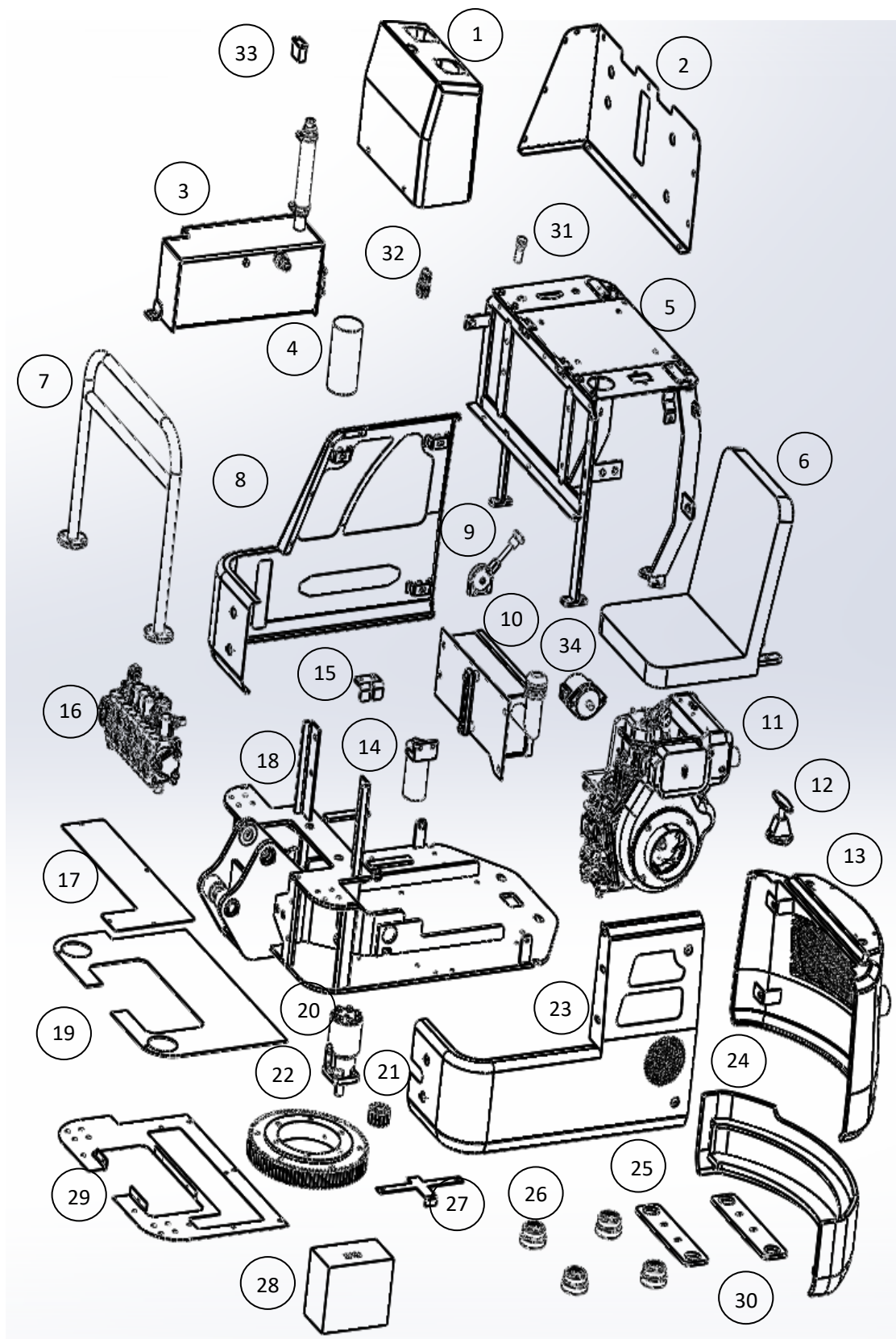
附件：整车零件 BOM 表



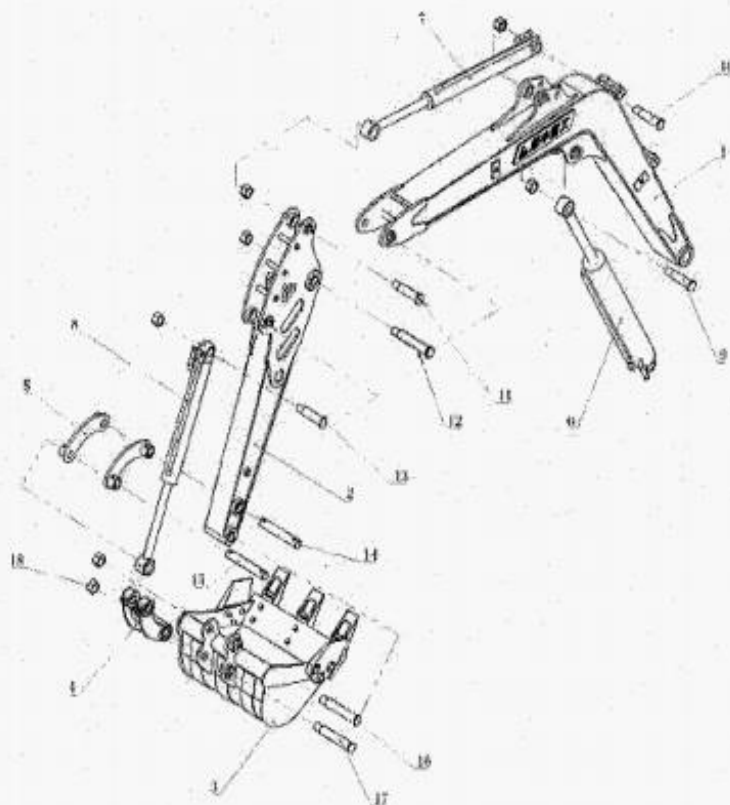
0.8T挖掘机总成爆炸图			
序号	名称	数量	备注
1	上车架总成	1	
2	下车架总成	1	
3	前装装部分总成	1	
4	回转支承总成	1	
5	动臂与上车架连接轴	1	
6	动臂缸与上车架连接轴	1	
7	锁轴套	2	



1. Lower chassis assembly			
No.	Part name	Quantity	Remarks
1	Blade Tilt Cylinder Pin	1	
2	Guiding Wheel Assembly	2	
3	Blade	1	
4	Track Assembly	2	
5	Blade Tilt Cylinder	1	
6	Blade Tilt Cylinder Pin 2	1	
7	Blade Pin	2	
8	Support Roller	4	
9	Central Swivel Joint	1	
10	Wheel Motor	2	
11	Drive Sprocket	2	

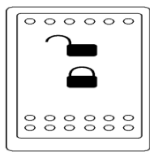
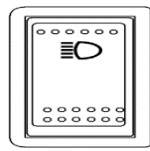
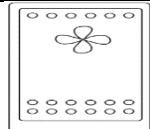
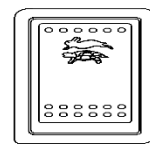


2. Upper Frame Assembly Bill of Materials (BOM)			
No.	Part name	Quantity	Remarks
1	Hood Panel Weldment	1	
2	Splash Plate	1	
3	Hydraulic Tank	1	
4	Return Oil Filter	1	
5	Seat Frame	1	
6	Seat	1	
7	Armrest	1	
8	Right Hood Weldment	1	
9	Throttle Assembly	1	
10	Diesel Tank	1	
11	Engine	1	
12	Main Power Switch	1	
13	Rear Hood Weldment	1	
14	Diesel Filter	1	
15	Swivel Joint Plate	1	
16	Multi-Way Valve	1	
17	Cover Plate	1	
18	Upper Frame Weldment	1	
19	Rubber Foot Mat	1	
20	Swivel Motor	1	
21	Swivel Bearing Small Gear	1	
22	Swivel Bearing Assembly	1	
23	Left Hood Weldment	1	
24	Counterweight	1	
25	Front Engine Bracket	1	
26	Engine Shock Pad	4	
27	Battery Clamp	1	
28	Battery	1	
29	Foot Pedal Weldment	1	
30	Rear Engine Bracket	1	
31	Oil Alarm	1	
32	Key Switch	1	
33	Timer	1	
34	Hydraulic Pump	1	
35	Wiring Harness Assembly	1	



3 前装部分				
序号	零件号	零件名称	数量	备注
1	3.1	动臂	1	
2	3.2	斗杆	1	
3	3.3	挖斗	1	
4	3.4	连杆	1	
5	3.5	推杆	1	
6	3.6	动臂油缸	1	
7	3.7	斗杆油缸	1	
8	3.8	挖斗油缸	1	
9	3.9	动臂中轴	1	
10	3.10	斗杆缸后轴	1	
11	3.11	斗杆缸前轴	1	
12	3.12	动臂前轴	1	
13	3.13	挖斗缸后轴	1	
14	3.14	斗杆中轴	1	
15	3.15	挖斗缸前轴	1	
16	3.16	斗杆前轴	1	
17	3.17	挖斗与连杆连接轴	1	
18	3.18	轴锁套	7	

Micro Excavator Electronic Control System Manual

1. Safety Lock Switch		1. When the safety switch is turned on, hydraulic operations can be performed. 2. When the safety switch is turned off, hydraulic operations will be disabled.
2. Light Switch		1. Press the switch to activate the headlights/work lights. 2. Release the switch to turn off the headlights/work lights.
3. Fan Switch		1. Press the switch to activate the fan. 2. Release the switch to turn off the fan.
4. High / Low Speed Switch		1. Press the switch to enter high-speed working mode while the excavator is moving. 2. Release the switch to enter low-speed working mode while the excavator is moving.

Function of the Buzzer

The function of the buzzer is to alert the driver of vehicle malfunctions and ensure that the vehicle is in normal working condition.

During operation or before starting work, the buzzer emits different beep sounds depending on the vehicle's functions. These sounds may indicate low engine oil pressure, low voltage, low fuel level, high engine temperature, engine malfunctions, and other warnings.

Buzzer sizes and shapes can vary significantly, but their functionality remains the same. In micro excavators equipped with electronic monitors, a small buzzer is embedded within the electronic monitor. For micro excavators without electronic monitors, the buzzer is typically fixed between the vehicle frames.◦

